

REQUIREMENTS TRACEABILITY MATRIX

CSI-100 Cell Sorting Instrument (IVD)

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Revision	D (supersedes Rev C)
Document type	Traceability matrix (DHF; ISO 13485 7.3, 21 CFR 820)
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Baseline	SRS Rev E; SRS-SW Rev F; RMF Rev D; software 3.0.3

Purpose

Bidirectional traceability from user needs through system and subsystem requirements, software requirements, design outputs, verification test cases and validation evidence, with risk controls, standards, labeling, anomalies, production controls and change impact. Generated from the controlled documents in Section 1.4.

Document Control

Rev	Date	Description of change	Author
A	Nov 2025	Initial matrix: user needs, system requirements, allocation.	A. Lodhi
B	Aug 2026	Added software requirements, design outputs, verification protocols and design transfer links (DTRR-1).	A. Lodhi
C	May 2027	V&V complete: verification and validation results, anomaly links, software architecture allocation, backward trace and completeness checks. Baseline for VVR-CSI-001 and the 510(k) submission.	A. Lodhi
D	Dec 2028	Baseline after CAPA-2028-014 (SRS-CSI-001 Rev E, SRS-SW-CSI-010 Rev F, RMF-CSI-001 Rev D, software 3.0.3). Added: subsystem requirement layer; test-case level trace; end-to-end thread per user need; risk trace at hazardous-situation level; standards, labeling and submission matrices; change impact and suspect-link record. Completeness checks extended from 16 to 27.	A. Lodhi

Summary of changes in Rev D

Change	Reason
New requirements SYS-327, FW-311, FW-312, FW-313, RA-107 traced up to user needs and hazards and down to design, tests and production controls	CAPA-2028-014 corrective actions CA1, CA6, CA8, PA11
Subsystem requirement layer (SR- <i>nnn.nn</i>) between system requirements and hardware design outputs	Audit observation AO-2028-11: hardware trace went directly from SYS to drawings, so subsystem specifications could not be shown complete
Verification traced to test case (DVP- <i>nnn TC-nn</i> ; STS- <i>x-nnn</i>), not only to protocol	Supports sampling by auditors and change impact analysis
Risk trace at hazardous-situation level (HS- <i>xx.y</i>) with control type	RMF-CSI-001 Rev D restructure
Standards, labeling and submission matrices	Reviewer requests during 510(k) review; EU IVDR technical documentation
Change impact and suspect-link record (Section 14)	Shows how the trace was kept valid through the CAPA change

Approvals

Function	Role	Signature / Date
Systems Engineering	Author, traceability owner	
Software Engineering	Software traceability	
V&V Engineering	Verification results	
Clinical Affairs	Validation results	
Quality Engineering	Design history file, risk management	
Regulatory Affairs	Submission	

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1. Introduction

1.1 Purpose

This matrix demonstrates that the CSI-100 design control chain is complete and consistent. It links every user need to the system requirements that address it and to the validation evidence that confirms it; every system requirement to its subsystem or software requirements, design outputs and verification test cases; and every risk control in the risk management file to the requirement that implements it and the evidence that it works. It also traces backward, so that every test, design output, production control and software item is justified by a requirement.

1.2 Scope

All CSI-100 configurations, options and accessories, software release 3.0.3 (Research Application 3.0.2, controller firmware 3.0.3, board FPGA 3.0.6) and the production baseline of DTP-CSI-001 as revised by CAPA-2028-014. Reagents and assays are traced in their own design history files.

1.3 Trace model

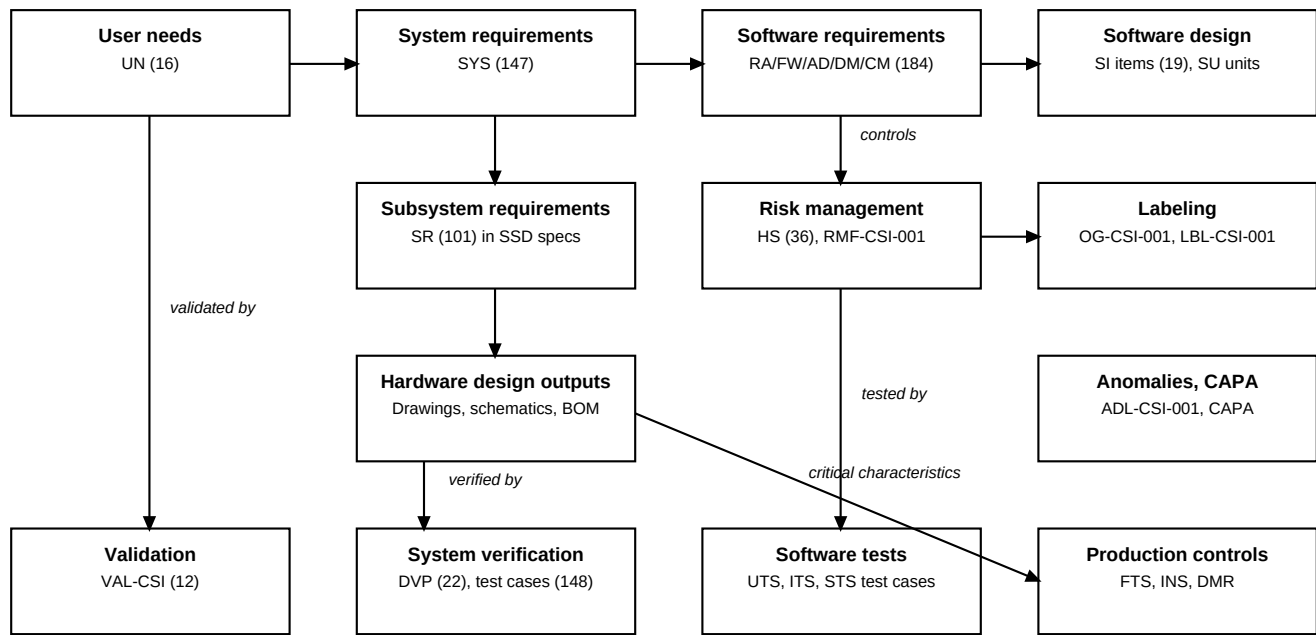


Figure 1. Trace model. Every arrow is traced in both directions in this document.

1.4 Source documents and baseline

Trace element	Source document (baseline)
User needs; system requirements with source, allocation and verification method	SRS-CSI-001 Rev E
Subsystem requirements and hardware design outputs	SSD-CSI-100 to SSD-CSI-950; drawings; BOM-CSI-0001 Rev D
Software requirements, parents, items, test level	SRS-SW-CSI-010 Rev F
Software items and units	SAD-SW-CSI-011 Rev C
Verification protocols and test cases; validation studies	VVP-CSI-001 Rev B; DVP-CSI-xxx protocols
Verification and validation results	PVR-CSI-001 with Add. 1, 2; SVR-CSI-001; SVR-SW-CSI-010 with Add. 1; DVAL-CSI-001
Hazardous situations and risk controls	RMF-CSI-001 Rev D (HA-CSI-001 Rev D)
Anomalies	ADL-CSI-001 Export 11
Production controls	DTP-CSI-001 Rev B; FTS-CSI-001 Rev B; INS-CSI-001 Rev B
Change records	CAPA-2028-014; ECO-0241, -0244, -0246; CR-SW-0512, -0514, -0521

The matrix is exported from the requirements management tool. For this revision every table and every completeness check in Section 2 was generated from the source documents above, so counts and links match those documents exactly.

1.5 Link types and rules

Link	From → to	Rule
derives from	SYS → UN, HA, standard	Every SYS has at least one source. Hazard sources name the hazard, not a situation.
allocated to	SYS → SS-01 to SS-10	Hardware allocation creates a subsystem requirement (SR); SS-10 allocation creates at least one software requirement.
refines	SR, RA/FW/AD/DM/CM → SYS	A child may refine more than one parent. A software requirement may also derive from a hazard (HA) directly.
implemented by	SR → drawing, schematic; SW req → SI item	Every design output is justified by at least one requirement.
verified by	SYS → DVP test case; SW req → UTS/ITS/STS	One primary test case per requirement; additional cases listed where a requirement is verified in more than one protocol.
validated by	UN → VAL study	Validation uses production-equivalent units and intended users; it is never cited as verification.
controls	Requirement → HS	Every risk control in RMF-CSI-001 cites the requirement that implements it.
suspect	Any link	Set automatically when either end changes; cleared only with an impact assessment (Section 14).

1.6 Conventions

- IDs marked **(D)** were added in this revision. Requirement numbers are never reused; RA-106 was assigned to a change request that was withdrawn (CR-SW-0498) and is not used.
- Result **Pass** means the requirement met its acceptance criterion in the cited report. **Pass (after ANM-nnnn)** means it passed after the anomaly was corrected and regression tested. **Deferred** anomalies are severity 4 and do not affect the result (ADL-CSI-001 Section 9).

- Test case IDs: DVP-nnn TC-nn is a system verification test case; STS-x-nnn a software system test case (x = part A to E of SRS-SW-CSI-010); ITS §n an integration test section; UTS-010-nn a unit test specification.
- Subsystem requirement IDs SR-nnn.nn refer to the subsystem design specification SSD-CSI-nnn, numbered in the order of their parent system requirements.

2. Completeness and Coverage Summary

These checks answer the questions auditors ask first: is every requirement tested, is every test justified by a requirement, and is every risk control implemented and verified? Each check is computed from the matrices in this document.

Area	Check	Items	Traced	Result
Forward	Every user need traces to at least one system requirement	16	16	Complete
Forward	Every user need traces to at least one validation study	16	16	Complete
Forward	Every system requirement has a source (user need, hazard, standard or regulation)	147	147	Complete
Forward	Every system requirement is allocated to a subsystem or the system	147	147	Complete
Forward	Every hardware-allocated system requirement has a subsystem requirement	82	82	Complete
Forward	Every system requirement allocated to software (SS-10) has a child software requirement	39	39	Complete
Forward	Every system requirement has a design output	147	147	Complete
Forward	Every system requirement has a verification method and test case	147	147	Complete
Forward	Every system requirement verified with result Pass	147	147	Complete
Forward	Every software requirement has a valid parent (SYS or HA)	184	184	Complete
Forward	Every software requirement is allocated to a software item in SAD-SW-CSI-011	184	184	Complete
Forward	Every software requirement has a test case at its test level	184	184	Complete
Forward	Every software requirement verified with result Pass	184	184	Complete
Risk	Every risk control requirement in RMF-CSI-001 Rev D exists in SRS-CSI-001 or SRS-SW-CSI-010	92	92	Complete
Risk	Every risk control requirement is verified with result Pass	92	92	Complete
Risk	Every hazardous situation has at least one traced risk control requirement	36	36	Complete
Risk	Every hazardous situation has implementation and effectiveness evidence	36	36	Complete
Risk	Every information-for-safety control traces to a labeling requirement (SYS-11xx or SYS-554)	17	17	Complete
Backward	Every verification protocol traces to at least one system requirement (no unjustified tests)	22	22	Complete
Backward	Every validation study traces to at least one user need	12	12	Complete
Backward	Every software item implements at least one software requirement	19	19	Complete
Backward	Every subsystem requirement has a parent system requirement	101	101	Complete
Backward	Every design output listed is justified by at least one system requirement	21	21	Complete
Backward	Every production critical characteristic traces to a design requirement	20	20	Complete
Backward	Every anomaly traces to a requirement or user need	32	32	Complete
Risk	Every DFMEA line traces to a hazardous situation in RMF-CSI-001	12	12	Complete
Change	Every requirement changed by CAPA-2028-014 re-verified and regression tested	5	5	Complete

During preparation of this revision the information-for-safety check found one gap: the IS control for HS-11.1 (installation by service only) traced to SYS-1203, an installation requirement, but not to a labeling requirement. RMF-CSI-001 Rev D was corrected before release to cite SYS-1104 (site preparation guide and service manual), which is verified by inspection in DVP-CSI-1101. The check is shown after the correction.

Conclusion: all 27 checks are complete; there are no orphan requirements, untested requirements, unjustified tests or unverified risk controls.

2.1 Coverage by level

Level	Count	Traced up	Traced down	Verified / validated
User needs	16	—	16 (100%)	16 validated (100%)
System requirements	147 (146 + 1 in Rev E)	147 (100%)	147 (100%)	147 verified (100%)
Subsystem requirements	101	101 (100%)	101 (100%)	Via parent test case and SSVR
Software requirements	184 (180 + 4 in Rev F)	184 (100%)	184 (100%)	184 tested (100%)
Hazardous situations	36	—	36 (100%)	36 controls verified (100%)
Risk control requirements	92	92 (100%)	—	92 verified (100%)

Verification methods for system requirements: T 72, I 35, A 12, D 31 (T = test, I = inspection, A = analysis, D = demonstration; a requirement may use more than one). Software test level: integration 57, system 92, unit 35. Every software requirement is also covered by a system test case.

3. Matrix A: User Needs to System Requirements and Validation

Forward trace from each user need to the system requirements that satisfy it (verified in Matrix B) and to the validation studies that confirm it. Verification and validation are in separate columns and are never used as evidence for each other.

User need	Statement	System requirements	Validation	Validation result
UN-01	Identify and enumerate cell populations in stained human specimens with results accurate ...	401, 402, 411, 421, 422, 433, 434, 443, 444, 445, 495, 632, 704, 806	VAL-01, VAL-04	Pass
UN-02	Physically separate target cell populations at high purity and recovery, with cells ...	461, 462, 464, 466, 631, 633, 634	VAL-02	Pass (after ANM-0056)
UN-03	Measure many markers on each cell at the same time (multicolor panels).	431, 432, 435, 441, 632	VAL-01	Pass
UN-04	Process specimens at a throughput that fits the laboratory workday.	451, 452, 453, 492, 494, 604	VAL-05	Pass
UN-05	Collect sorted cells into tubes, multiwell plates and slides, including single-cell ...	464, 481, 482, 483, 484, 485, 631	VAL-02	Pass (after ANM-0056)
UN-06	Operate the instrument without risk of injury from laser light, high voltage, moving ...	551, 803, 804, 902, 1304	VAL-03	Pass (after ANM-0060)
UN-07	Install the instrument in a standard clinical laboratory without special facilities.	301, 302, 303, 304, 305, 306, 311, 312, 313, 321, 322, 324, 325, 326, 327 (D), 331, 438, 1104, 1201, 1203, 1204	VAL-06	Pass
UN-08	Obtain consistent performance from day to day with minimal user adjustment, confirmed by ...	436, 437, 444, 465, 468, 613, 621, 622, 624, 806, 1201	VAL-04	Pass
UN-09	Set up, run, clean and shut down the instrument through guided, largely automated ...	437, 465, 492, 493, 601, 602, 603, 611, 612, 614, 805, 901	VAL-03	Pass (after ANM-0060)
UN-10	Keep results and records secure, attributable and traceable, with controlled user access.	635, 701, 702, 703, 705, 706, 709, 802, 904, 1007	VAL-09	Pass (after ANM-0058)
UN-11	Keep samples and sorted cells at a chosen temperature during the run.	4101, 4102, 803	VAL-08	Pass
UN-12	Keep the instrument in service with predictable maintenance and fast repair.	1001, 1002, 1003, 1004, 1005, 1006, 1007, 1104, 1202	VAL-10	Pass
UN-13	Use the instrument in the United States and the European Union in compliance with IVD ...	1101, 1102, 1105, 1301, 1302, 1303, 1304, 1305, 1306	VAL-12	Pass
UN-14	Avoid carryover and cross-contamination between specimens.	471, 472, 603	VAL-07	Pass
UN-15	Handle cells of different sizes and fragility.	423, 463, 467, 491, 611	VAL-02	Pass (after ANM-0056)
UN-16	Choose the laser and detector configuration that fits the laboratory's assays, and ...	431, 432, 442, 623, 1008	VAL-11	Pass

System requirement numbers shown without the SYS- prefix. Validation results from DVAL-CSI-001. UN-08 precision validation (VAL-CSI-04) was re-confirmed after CAPA-2028-014 closed ANM-0062.

4. Matrix A2: End-to-End Thread by User Need

One row per user need showing the complete chain, for a reviewer who wants to follow one need end to end. Prefixes SS-, SI-, VAL-CSI- and HS- are omitted. Hazardous situations are those controlled by the need's system requirements or their software children.

UN	Need (abbreviated)	SYS	Subsystems	SR	SW reqs / items	Verification protocols	Validation	Hazardous situations	Status
UN-01	Identify and enumerate cell populations in stained human ...	14	02, 03, 04, 05, 06, 10	18	20 / 02, 04, 05, 11, 12, 21, 22, 32	DVP-401, 402, 403, 408, 601, 701, 801	01, 04	08.4, 08.5, 08.6	Complete
UN-02	Physically separate target cell populations at high ...	7	06, 07, 10	4	11 / 01, 02, 03, 11, 22	DVP-405, 601	02	07.3, 15.2	Complete
UN-03	Measure many markers on each cell at the same time ...	5	04, 05, 06, 10	5	5 / 02, 12, 21	DVP-402, 403, 601	01	—	Complete
UN-04	Process specimens at a throughput that fits the ...	6	01, 02, 06	5	8 / 01, 02, 04, 10, 11, 21, 41	DVP-404, 408, 601	05	04.3, 08.6	Complete
UN-05	Collect sorted cells into tubes, multiwell plates and ...	7	06, 07, 08, 10	5	12 / 02, 03, 10, 11, 12, 21, 22, 31	DVP-405, 407, 601	02	—	Complete
UN-06	Operate the instrument without risk of injury from laser ...	5	01, 08, 09	4	2 / 01, 04	DVP-504, 801, 901, 1301	03	04.1	Complete
UN-07	Install the instrument in a standard clinical laboratory ...	21	01, 04, 05, 06, 10	10	3 / 01, 05, 10, 12	DVP-301, 403, 1101, 1201	06	01.2, 08.2, 11.1, 17.2	Complete
UN-08	Obtain consistent performance from day to day with ...	11	03, 04, 06, 07, 10	6	20 / 01, 03, 05, 11, 13, 20, 21, 22, 40	DVP-402, 403, 405, 601, 801, 1201	04	01.2, 07.1, 07.2, 08.1, 08.2, 08.3, 13.1, 17.2	Complete
UN-09	Set up, run, clean and shut down the instrument through ...	12	01, 04, 07, 10	11	22 / 01, 03, 04, 10, 13, 14, 21, 22, 32	DVP-403, 405, 408, 601, 801, 901	03	04.4, 07.2, 15.1	Complete
UN-10	Keep results and records secure, attributable and ...	10	06, 10	2	50 / 01, 02, 03, 05, 10, 12, 20, 21, 22, 30, 31, 32, 40, 41	DVP-601, 701, 801, 901, 1001	09	01.2, 07.1, 08.4, 15.2, 16.1, 16.2, 17.1	Complete
UN-11	Keep samples and sorted cells at a chosen temperature ...	3	01, 02, 08, 09	4	4 / 04, 21	DVP-408, 801	08	—	Complete
UN-12	Keep the instrument in service with predictable ...	9	01, 06, 10	2	16 / 02, 05, 10, 12, 21, 31	DVP-1001, 1101, 1201	10	01.2, 11.1	Complete
UN-13	Use the instrument in the United States and the European ...	9	10	0	3 / 01, 21, 41	DVP-1101, 1301	12	01.3, 04.3, 05.1, 06.2, 07.1, 07.3, 08.5, 09.2, 17.1	Complete
UN-14	Avoid carryover and cross-contamination between specimens.	3	01, 02, 10	3	2 / 04, 10	DVP-406, 601	07	04.3, 04.4	Complete
UN-15	Handle cells of different sizes and fragility.	5	01, 03, 05, 07, 10	6	9 / 03, 12, 14, 20, 21, 22, 32	DVP-402, 405, 408, 601	02	09.1	Complete
UN-16	Choose the laser and detector configuration that fits the ...	5	04, 05, 10	5	9 / 05, 10, 12, 20, 32	DVP-402, 403, 601, 1001	11	08.1	Complete

5. Matrix B: System Requirements to Design and Verification

One row per system requirement in SRS-CSI-001 Rev E, grouped by SRS section. Children are subsystem requirements for hardware allocations and software requirements for SS-10 allocations. Requirement text is abbreviated; the SRS is the controlling text.

ID	Requirement (abbreviated)	Source	Alloc.	Children (SR / SW)	Design output	Verification (method: test case)	Report	Risk (HS)	Result
SRS-CSI-001 section 3									
SYS-301	The sorter shall not exceed 71 cm height, 122 cm width and 71 cm depth (28 × 48 × 28 in), excluding ...	UN-07	SYS	—	DWG-CSI-0001; SCH-CSI-6000	I: DVP-301 TC-01	PVR-CSI-001	—	Pass
SYS-302	The sorter shall weigh no more than 185 kg (408 lb) in its heaviest configuration.	UN-07	SYS	—	DWG-CSI-0001; SCH-CSI-6000	I: DVP-301 TC-02	PVR-CSI-001	—	Pass
SYS-303	The fluidics supply module shall not exceed 66 cm height, 81 cm width and 66 cm depth (26 × 32 × 26 ...	UN-07	01	SR-100.01	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100	I: DVP-301 TC-03	PVR-CSI-001	—	Pass
SYS-304	The fluidics supply module shall be mobile on lockable casters with leveling feet.	UN-07	01	SR-100.02	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100	I: DVP-301 TC-04	PVR-CSI-001	—	Pass
SYS-305	The sorter shall operate on a laboratory bench or on an optional bench no larger than 85.1 × 266.7 ...	UN-07	SYS	—	DWG-CSI-0001; SCH-CSI-6000	I: DVP-301 TC-05	PVR-CSI-001	—	Pass
SYS-306	The umbilical between sorter and fluidics supply module shall allow separation of up to 2.7 m (9 ...	UN-07	01	SR-100.03	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100	I: DVP-301 TC-06	PVR-CSI-001	—	Pass
SYS-311	The system shall operate from 100, 115 or 230 VAC (±10%), 50 or 60 Hz, single phase.	UN-07	06	SR-600.01	SSD-CSI-600; SCH-CSI-6000	T: DVP-301 TC-07	PVR-CSI-001	—	Pass
SYS-312	Total power consumption of the sorter and fluidics supply module shall not exceed 1,500 W.	UN-07	SYS	—	DWG-CSI-0001; SCH-CSI-6000	T: DVP-301 TC-08	PVR-CSI-001	—	Pass
SYS-313	The system shall require no more than one 20 A outlet for the sorter and fluidics supply module ...	UN-07	SYS	—	DWG-CSI-0001; SCH-CSI-6000	A: DVP-301 TC-09	PVR-CSI-001	—	Pass
SYS-314	The fluidics supply module shall be powered and switched only from the sorter.	HA-12	01, 06	SR-100.04, SR-600.02	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100; SSD-CSI-600; SCH-CSI-6000	I: DVP-301 TC-10	PVR-CSI-001	—	Pass
SYS-315	The system shall be protected by a resettable circuit breaker.	HA-12	06	SR-600.03	SSD-CSI-600; SCH-CSI-6000	I: DVP-301 TC-11	PVR-CSI-001	12.1	Pass

ID	Requirement (abbreviated)	Source	Alloc.	Children (SR / SW)	Design output	Verification (method: test case)	Report	Risk (HS)	Result
SYS-316	The system shall meet installation category II and pollution degree 2 as defined in IEC 61010-1.	IEC 61010-1	SYS	—	DWG-CSI-0001; SCH-CSI-6000	A: DVP-301 TC-12	PVR-CSI-001	—	Pass
SYS-321	The system shall meet all performance requirements at ambient temperatures of 15 to 30 °C.	UN-07	SYS	—	DWG-CSI-0001; SCH-CSI-6000	T: DVP-301 TC-13	PVR-CSI-001	—	Pass
SYS-322	The system shall meet all performance requirements at 5 to 80% relative humidity, non-condensing.	UN-07	SYS	—	DWG-CSI-0001; SCH-CSI-6000	T: DVP-301 TC-14	PVR-CSI-001	—	Pass
SYS-323	The system shall operate at altitudes up to 2,000 m.	IEC 61010-1	SYS	—	DWG-CSI-0001; SCH-CSI-6000	A: DVP-301 TC-15	PVR-CSI-001	—	Pass
SYS-324	The system shall withstand storage at 1 to 40 °C without loss of performance.	UN-07	SYS	—	DWG-CSI-0001; SCH-CSI-6000	T: DVP-301 TC-16	PVR-CSI-001	—	Pass
SYS-325	The system shall require no special room facilities, no external compressed air and no external ...	UN-07	01	SR-100.05	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100	D: DVP-301 TC-17	PVR-CSI-001	—	Pass
SYS-326	The system shall accept optional facility air at 80 to 100 PSI, dry and oil-free.	UN-07	01	SR-100.06	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100	T: DVP-301 TC-18	PVR-CSI-001	—	Pass
SYS-327 (D)	Every temperature-sensitive component (lasers, PMTs, cameras, electronics) shall stay within its ...	UN-07, HA-08	04, 05, 06, 10	SR-400.15, SR-500.10, SR-600.18, RA-107 (D), FW-311 (D), FW-312 (D)	SSD-CSI-400; DWG-CSI-4000; SSD-CSI-500; DWG-CSI-5100, DWG-CSI-5200; SSD-CSI-600; SCH-CSI-6000; SAD-SW-CSI-011	T, A: DVP-301 TC-20; DVP-403 TC-09	PVR-CSI-001 Add. 2	08.2, 17.2	Pass (PVR Add. 2)
SYS-331	Acoustic noise at the operator position shall not exceed 62 dBA during sorting.	UN-07	SYS	—	DWG-CSI-0001; SCH-CSI-6000	T: DVP-301 TC-19	PVR-CSI-001	—	Pass
SRS-CSI-001 section 4									
SYS-401	The coefficient of variation (CV) of the full G0/G1 peak for propidium iodide-stained chicken ...	UN-01	03, 04, 05	SR-300.01, SR-400.01, SR-500.01, AD-304	SSD-CSI-300; DWG-CSI-3100; SSD-CSI-400; DWG-CSI-4000; SSD-CSI-500; DWG-CSI-5100, DWG-CSI-5200	T: DVP-401 TC-01	PVR-CSI-001	—	Pass (after ANM-0012)

ID	Requirement (abbreviated)	Source	Alloc.	Children (SR / SW)	Design output	Verification (method: test case)	Report	Risk (HS)	Result
SYS-402	The CV of the full G0/G1 peak for Hoechst-stained chicken erythrocyte nuclei, excited at 405 nm or ...	UN-01	03, 04, 05	SR-300.02, SR-400.02, SR-500.02, AD-304	SSD-CSI-300; DWG-CSI-3100; SSD-CSI-400; DWG-CSI-4000; SSD-CSI-500; DWG-CSI-5100, DWG-CSI-5200	T: DVP-401 TC-02	PVR-CSI-001	—	Pass
SYS-411	The doublet-to-singlet ratio for chicken erythrocyte nuclei shall be 1.95 to 2.05, for PI excited ...	UN-01	05, 06	SR-500.03, SR-600.04, AD-303, AD-304	SSD-CSI-500; DWG-CSI-5100, DWG-CSI-5200; SSD-CSI-600; SCH-CSI-6000	T: DVP-401 TC-03	PVR-CSI-001	—	Pass
SYS-421	Forward and side scatter shall resolve 0.5 μ m beads from instrument noise.	UN-01	04, 05	SR-400.03, SR-500.04	SSD-CSI-400; DWG-CSI-4000; SSD-CSI-500; DWG-CSI-5100, DWG-CSI-5200	T: DVP-402 TC-01	PVR-CSI-001	—	Pass
SYS-422	Scatter sensitivity shall allow separation of fixed platelets from noise and identification of ...	UN-01	04, 05	SR-400.04, SR-500.05	SSD-CSI-400; DWG-CSI-4000; SSD-CSI-500; DWG-CSI-5100, DWG-CSI-5200	T: DVP-402 TC-02	PVR-CSI-001	—	Pass
SYS-423	The forward scatter path shall accept interchangeable neutral density filters of 0.5, 1.0, 1.5 and ...	UN-15	05	SR-500.06, AD-305	SSD-CSI-500; DWG-CSI-5100, DWG-CSI-5200	I: DVP-402 TC-03	PVR-CSI-001	—	Pass
SYS-431	The system shall support up to six installed lasers, with up to four active at the same time.	UN-03, UN-16	04	SR-400.05	SSD-CSI-400; DWG-CSI-4000	D: DVP-403 TC-01	PVR-CSI-001	—	Pass
SYS-432	The 488 nm and 633 nm lasers shall be standard in every configuration. 375, 405, 445 and 561 nm ...	UN-03, UN-16	04	SR-400.06	SSD-CSI-400; DWG-CSI-4000	I: DVP-403 TC-02	PVR-CSI-001	—	Pass
SYS-433	Minimum laser power out of the laser head shall be: 375 nm > 7 mW; 405 nm > 50 mW; 445 nm > 20 mW ...	UN-01	04	SR-400.07, FW-305, AD-201, AD-202, AD-203	SSD-CSI-400; DWG-CSI-4000	T: DVP-403 TC-03	PVR-CSI-001	—	Pass
SYS-434	Elliptical beams at the interrogation point shall measure 9 ± 3 μ m in height and 65 ± 7 μ m in width.	UN-01	04	SR-400.08	SSD-CSI-400; DWG-CSI-4000	T: DVP-403 TC-04	PVR-CSI-001	—	Pass
SYS-435	Each active laser shall be focused on a spatially separated beam spot. Lasers sharing a spot ...	UN-03	04, 06	SR-400.09, SR-600.05, FW-305, FW-306	SSD-CSI-400; DWG-CSI-4000; SSD-CSI-600; SCH-CSI-6000	T: DVP-403 TC-05	PVR-CSI-001	—	Pass

ID	Requirement (abbreviated)	Source	Alloc.	Children (SR / SW)	Design output	Verification (method: test case)	Report	Risk (HS)	Result
SYS-436	Optical alignment shall be fixed at the factory and shall require no user adjustment.	UN-08	04	SR-400.10, AD-301	SSD-CSI-400; DWG-CSI-4000	D: DVP-403 TC-06	PVR-CSI-001	01.2	Pass
SYS-437	The system shall reach full performance within 30 minutes of laser power-on.	UN-08, UN-09	04	SR-400.11, AD-204	SSD-CSI-400; DWG-CSI-4000	T: DVP-403 TC-07	PVR-CSI-001	—	Pass
SYS-438	Lasers shall be air-cooled and shall need no external cooling or special power.	UN-07	04	SR-400.12	SSD-CSI-400; DWG-CSI-4000	I: DVP-403 TC-08	PVR-CSI-001	—	Pass
SYS-441	The system shall measure forward scatter, side scatter and at least 18 fluorescence parameters ...	UN-03	05	SR-500.07, FW-304, AD-302	SSD-CSI-500; DWG-CSI-5100, DWG-CSI-5200	T: DVP-402 TC-04	PVR-CSI-001	—	Pass
SYS-442	Each fluorescence detector shall have a user-replaceable bandpass filter and each detector array ...	UN-16	05	SR-500.08	SSD-CSI-500; DWG-CSI-5100, DWG-CSI-5200	I: DVP-402 TC-05	PVR-CSI-001	—	Pass
SYS-443	The system shall report height, area and width for every detected parameter.	UN-01	06	SR-600.06, RA-301, FW-201, FW-303	SSD-CSI-600; SCH-CSI-6000	T: DVP-402 TC-06	PVR-CSI-001	—	Pass
SYS-444	Signals from each beam spot shall be time-aligned to the same event, with inter-laser delay set ...	UN-01, UN-08	06, 10	SR-600.07, RA-802, FW-201, AD-205	SSD-CSI-600; SCH-CSI-6000; SAD-SW-CSI-011	T: DVP-402 TC-07	PVR-CSI-001	—	Pass
SYS-445	Data shall be digitized with a linear dynamic range of at least 18 bits.	UN-01	06	SR-600.08, RA-302, FW-202	SSD-CSI-600; SCH-CSI-6000	A: DVP-402 TC-08	PVR-CSI-001	—	Pass
SYS-451	The system shall acquire at least 70,000 events per second with 8 parameters and 12 compensation ...	UN-04	06	SR-600.09, RA-303, RA-304, FW-202, FW-806	SSD-CSI-600; SCH-CSI-6000	T: DVP-404 TC-01	PVR-CSI-001	08.6	Pass
SYS-452	The system shall acquire more than 100,000 events per second when fewer parameters are recorded.	UN-04	06	SR-600.10, RA-303, FW-202, FW-806	SSD-CSI-600; SCH-CSI-6000	T: DVP-404 TC-02	PVR-CSI-001	—	Pass
SYS-453	Sample flow rate shall be user-selectable over at least 10 to 80 μ L/min.	UN-04	02	SR-100.07, RA-206, AD-402	SSD-CSI-100; DWG-CSI-0001	T: DVP-404 TC-03	PVR-CSI-001	—	Pass
SYS-461	For a four-way sort at 70 PSI, 90 kHz drop drive and an average threshold rate of 25,000 events per ...	UN-02	06, 07	SR-600.11, SR-700.01, FW-204	SSD-CSI-600; SCH-CSI-6000; SSD-CSI-700; DWG-CSI-7000	T: DVP-405 TC-01	PVR-CSI-001	—	Pass (after ANM-0003, ANM-0047)
SYS-462	The system shall sort into up to four populations at the same time.	UN-02	07	SR-700.02	SSD-CSI-700; DWG-CSI-7000	D: DVP-405 TC-02	PVR-CSI-001	—	Pass
SYS-463	Drop drive frequency shall be adjustable from 1 to 100 kHz or wider.	UN-15	07	SR-700.03, RA-405, AD-502	SSD-CSI-700; DWG-CSI-7000	T: DVP-405 TC-03	PVR-CSI-001	—	Pass
SYS-464	The system shall resolve particle position within the drop to 1/32 of a drop and shall provide ...	UN-02, UN-05	06, 10	SR-600.12, RA-604, RA-605, FW-203, AD-703	SSD-CSI-600; SCH-CSI-6000; SAD-SW-CSI-011	T: DVP-405 TC-04	PVR-CSI-001	—	Pass

ID	Requirement (abbreviated)	Source	Alloc.	Children (SR / SW)	Design output	Verification (method: test case)	Report	Risk (HS)	Result
SYS-465	The system shall automatically determine drop delay using fluorescent drop-delay beads.	UN-08, UN-09	07, 10	SR-700.04, RA-407	SSD-CSI-700; DWG-CSI-7000; SAD-SW-CSI-011	T: DVP-405 TC-05	PVR-CSI-001	07.2	Pass
SYS-466	Sorted cell lines and human PBMCs shall remain viable and proliferate after sorting at every ...	UN-02	SYS	—	DWG-CSI-0001	T: DVP-405 TC-06	PVR-CSI-001	—	Pass
SYS-467	The system shall provide nozzles of 70, 85 and 100 µm, with a 130 µm nozzle as an option, each with ...	UN-15	03, 07	SR-300.03, SR-700.05	SSD-CSI-300; DWG-CSI-3100; SSD-CSI-700; DWG-CSI-7000	I: DVP-405 TC-07	PVR-CSI-001	—	Pass
SYS-468	Changing the nozzle shall not require optical realignment.	UN-08	03	SR-300.04, AD-503	SSD-CSI-300; DWG-CSI-3100	D: DVP-405 TC-08	PVR-CSI-001	—	Pass
SYS-471	Sample carryover from one tube to the next shall not exceed 0.5% after the automatic sample line ...	UN-14	02	SR-100.08	SSD-CSI-100; DWG-CSI-0001	T: DVP-406 TC-01	PVR-CSI-001	—	Pass (after ANM-0019)
SYS-472	The system shall flush the inside and outside of the sample line with sheath after every tube and ...	UN-14	02, 10	SR-100.09, RA-205	SSD-CSI-100; DWG-CSI-0001; SAD-SW-CSI-011	D: DVP-406 TC-02	PVR-CSI-001	04.3	Pass
SYS-481	The system shall provide two-way sorting into 12 × 75 mm and 15 mL tubes, and four-way sorting into ...	UN-05	08	SR-800.01, RA-601	SSD-CSI-800; DWG-CSI-0001	D: DVP-407 TC-01	PVR-CSI-001	—	Pass
SYS-482	With the plate deposition option, the system shall sort into 6-, 24-, 48-, 96- and 384-well plates ...	UN-05	08	SR-800.02, RA-601, FW-208	SSD-CSI-800; DWG-CSI-0001	D: DVP-407 TC-02	PVR-CSI-001	—	Pass
SYS-483	With the plate deposition option, the system shall deposit single cells so that at least 95% of ...	UN-05	07, 08	SR-700.06, SR-800.03, FW-206, FW-208, AD-504	SSD-CSI-700; DWG-CSI-7000; SSD-CSI-800; DWG-CSI-0001	T: DVP-407 TC-03	PVR-CSI-001	—	Pass (after ANM-0037)
SYS-484	The system shall record the measured parameters of each cell deposited in a plate well (index ...	UN-05	10	RA-705	SAD-SW-CSI-011	D: DVP-407 TC-04	PVR-CSI-001	—	Pass
SYS-485	The system shall allow users to define custom collection devices.	UN-05	10	RA-601, AD-702	SAD-SW-CSI-011	D: DVP-407 TC-05	PVR-CSI-001	—	Pass
SYS-491	Sheath pressure shall be software-controlled from 5 to 75 PSI.	UN-15	01	SR-100.10, FW-401, AD-401, AD-404	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100	T: DVP-408 TC-01	PVR-CSI-001	09.1	Pass
SYS-492	The fluidics supply module shall hold at least 10 L of sheath, 10 L of waste, 5 L of shutdown ...	UN-04, UN-09	01	SR-100.11	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100	I: DVP-408 TC-02	PVR-CSI-001	—	Pass

ID	Requirement (abbreviated)	Source	Alloc.	Children (SR / SW)	Design output	Verification (method: test case)	Report	Risk (HS)	Result
SYS-493	The system shall warn the user before sheath runs out or waste is full, and shall prevent sorting ...	UN-09, HA-15	01, 10	SR-100.12, RA-103, RA-207, FW-402, AD-403	SSD-CSI-100; DWG-CSI-1000; DWG-CSI-1100; SAD-SW-CSI-011	T: DVP-408 TC-03	PVR-CSI-001	15.1	Pass
SYS-494	The sample chamber shall accept 15 mL, 12 × 75 mm and 1 mL tubes.	UN-04	02	SR-100.13, RA-205	SSD-CSI-100; DWG-CSI-0001	I: DVP-408 TC-04	PVR-CSI-001	04.3	Pass
SYS-495	The system shall agitate the sample tube at user-selectable speeds.	UN-01	02	SR-100.14, RA-206	SSD-CSI-100; DWG-CSI-0001	D: DVP-408 TC-05	PVR-CSI-001	—	Pass
SYS-4101	The sample chamber shall hold a user-selected temperature, reached within 45 minutes with the ...	UN-11	02	SR-100.15, RA-206, AD-405	SSD-CSI-100; DWG-CSI-0001	T: DVP-408 TC-06	PVR-CSI-001	—	Pass (after ANM-0039)
SYS-4102	With the temperature control option, collection tubes shall be held at a user-selected temperature ...	UN-11	08	SR-800.04, AD-505	SSD-CSI-800; DWG-CSI-0001	T: DVP-408 TC-07	PVR-CSI-001	—	Pass
SRS-CSI-001 section 5									
SYS-501	The system shall comply with IEC 61010-1 and IEC 61010-2-101 for protection against electric shock ...	HA-12, IEC 61010	SYS	—	DWG-CSI-0001; SCH-CSI-6000	T: DVP-501 TC-01	SVR-CSI-001	02.2, 12.1, 12.2	Pass
SYS-502	Protective earth continuity, dielectric strength and touch current shall be tested on every unit in ...	HA-12	SYS	—	DWG-CSI-0001; SCH-CSI-6000	T: DVP-501 TC-02	SVR-CSI-001	02.2, 12.1	Pass
SYS-503	The system shall comply with IEC 61326-1 and IEC 61326-2-6 for electromagnetic compatibility.	HA-13, IEC 61326	SYS	—	DWG-CSI-0001; SCH-CSI-6000	T: DVP-501 TC-03	SVR-CSI-001	13.1	Pass (after ANM-0021)
SYS-511	The system shall be a Class 1 laser product per IEC 60825-1 in all user-accessible conditions.	HA-01, IEC 60825-1	SYS	—	DWG-CSI-0001; SCH-CSI-6000	T: DVP-502 TC-01	SVR-CSI-001	01.1, 01.3	Pass
SYS-512	Opening the flow cell access door shall block all laser beams by a shutter before any beam is ...	HA-01	04	SR-400.13, FW-502	SSD-CSI-400; DWG-CSI-4000	T: DVP-502 TC-02	SVR-CSI-001	01.1, 05.1	Pass
SYS-513	Laser warning labels shall be placed on all service-access panels behind which Class 3B radiation ...	HA-01	SYS	—	DWG-CSI-0001; SCH-CSI-6000	I: DVP-502 TC-03	SVR-CSI-001	01.1	Pass (after ANM-0032)
SYS-521	The sort collection chamber door shall not open unless the sort block door is closed.	HA-05	07	SR-700.07, FW-502	SSD-CSI-700; DWG-CSI-7000	T: DVP-503 TC-01	PVR-CSI-001	01.1, 05.1	Pass
SYS-522	An illuminated warning light and a software indicator shall show whenever deflection plate voltage ...	HA-02	07, 10	SR-700.08, RA-103, RA-1003, FW-608	SSD-CSI-700; DWG-CSI-7000; SAD-SW-CSI-011	T: DVP-503 TC-02	PVR-CSI-001	02.1, 02.3	Pass
SYS-523	Deflection plate voltage shall not exceed 12,000 V between plates and shall be insulated so that no ...	HA-02	07	SR-700.09, FW-307, AD-502	SSD-CSI-700; DWG-CSI-7000	T: DVP-503 TC-03	PVR-CSI-001	02.1, 02.2	Pass

ID	Requirement (abbreviated)	Source	Alloc.	Children (SR / SW)	Design output	Verification (method: test case)	Report	Risk (HS)	Result
SYS-524	The system shall fully depressurize within 10 seconds of main power switch-off.	HA-09	01	SR-100.16, FW-403	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100	T: DVP-503 TC-04	PVR-CSI-001	09.2	Pass
SYS-531	The system shall provide an illuminated emergency stop button next to the sample loading port.	HA-06	06	SR-600.13	SSD-CSI-600; SCH-CSI-6000	I: DVP-503 TC-05	PVR-CSI-001	06.1	Pass
SYS-532	Pressing the emergency stop shall, within 2 seconds: unload the sample tube, stop plate stage ...	HA-02, HA-06	06, 10	SR-600.14, FW-601	SSD-CSI-600; SCH-CSI-6000; SAD-SW-CSI-011	T: DVP-503 TC-06	PVR-CSI-001	02.1, 06.2	Pass
SYS-533	The emergency stop button shall light while the sample loading port is moving.	HA-06	06	SR-600.15, FW-501	SSD-CSI-600; SCH-CSI-6000	T: DVP-503 TC-07	PVR-CSI-001	06.1	Pass
SYS-541	The system shall monitor the drop breakoff continuously during sorting.	HA-04, HA-07	07, 10	SR-700.10, FW-210	SSD-CSI-700; DWG-CSI-7000; SAD-SW-CSI-011	T: DVP-503 TC-08	PVR-CSI-001	04.2	Pass
SYS-542	If the breakoff moves outside its tolerance, the system shall pause sorting until the breakoff ...	HA-07	07, 10	SR-700.11, FW-205	SSD-CSI-700; DWG-CSI-7000; SAD-SW-CSI-011	T: DVP-503 TC-09	PVR-CSI-001	07.1	Pass
SYS-543	If a nozzle clog is detected, the system shall, within 2 seconds: stop the stream, turn off plate ...	HA-04	07, 10	SR-700.12, FW-602	SSD-CSI-700; DWG-CSI-7000; SAD-SW-CSI-011	T: DVP-503 TC-10	PVR-CSI-001	04.2, 15.2	Pass (after ANM-0011)
SYS-551	With the sort block and collection chamber doors closed and the aerosol evacuation option running ...	HA-04, UN-06	09	SR-900.01, RA-208	SSD-CSI-900; DWG-CSI-0001	T: DVP-504 TC-01	PVR-CSI-001	04.1	Pass
SYS-552	The aerosol evacuation option shall provide at least 7 CFM in normal mode and at least 30 CFM in ...	HA-04	09	SR-900.02	SSD-CSI-900; DWG-CSI-0001	T: DVP-504 TC-02	PVR-CSI-001	04.1	Pass
SYS-553	All wetted surfaces and surfaces exposed to specimens shall withstand the decontamination agents ...	HA-10	SYS	—	DWG-CSI-0001; SCH-CSI-6000	T: DVP-504 TC-03	PVR-CSI-001	10.1	Pass
SYS-554	The IFU shall describe safe handling of a clog and decontamination after sorting infectious ...	HA-04	SYS	RA-504	DWG-CSI-0001; SCH-CSI-6000	I: DVP-504 TC-04	PVR-CSI-001	04.1, 04.2, 04.4	Pass
SYS-561	Pressurized tanks shall be rated for at least 1.5 times the maximum operating pressure and shall ...	HA-09	01	SR-100.17	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100	A, T: DVP-505 TC-01	PVR-CSI-001	09.1	Pass
SYS-562	Moving parts (loading port, plate stage, aspirator drawer) shall not exert a pinching force greater ...	HA-06	02, 08	SR-100.18, SR-800.05, FW-309	SSD-CSI-100; DWG-CSI-0001; SSD-CSI-800	T: DVP-505 TC-02	PVR-CSI-001	06.1	Pass
SYS-563	The system shall remain stable when tilted 10° in any direction.	HA-11, IEC 61010-1	SYS	—	DWG-CSI-0001; SCH-CSI-6000	T: DVP-505 TC-03	PVR-CSI-001	11.1	Pass

ID	Requirement (abbreviated)	Source	Alloc.	Children (SR / SW)	Design output	Verification (method: test case)	Report	Risk (HS)	Result
SRS-CSI-001 section 6									
SYS-601	The system shall perform an automated fluidics startup that primes all fluid lines and removes air.	UN-09	01, 10	SR-100.19, RA-201	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100; SAD-SW-CSI-011	D: DVP-601 TC-01	PVR-CSI-001	—	Pass
SYS-602	The system shall perform an automated fluidics shutdown that leaves the flow path filled with 70% ...	UN-09	01, 10	SR-100.20, RA-202	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100; SAD-SW-CSI-011	D: DVP-601 TC-02	PVR-CSI-001	—	Pass
SYS-603	The system shall provide automated cleaning modes: flow cell clean, sample line backflush, prime ...	UN-09, UN-14	01, 10	SR-100.21, RA-203	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100; SAD-SW-CSI-011	D: DVP-601 TC-03	PVR-CSI-001	04.4	Pass
SYS-604	The system shall be ready to run daily QC within 45 minutes of power-on, including laser warm-up.	UN-04	SYS	CM-302	DWG-CSI-0001	T: DVP-601 TC-04	PVR-CSI-001	—	Pass
SYS-611	Selecting a nozzle size in software shall load a verified default setup for sheath pressure, drop ...	UN-09, UN-15	07, 10	SR-700.13, RA-401, RA-403, RA-405, AD-701	SSD-CSI-700; DWG-CSI-7000; SAD-SW-CSI-011	D: DVP-601 TC-05	PVR-CSI-001	—	Pass
SYS-612	The system shall display a live image of the drop breakoff and of the side streams.	UN-09	07	SR-700.14, RA-404, FW-209, AD-501	SSD-CSI-700; DWG-CSI-7000	D: DVP-601 TC-06	PVR-CSI-001	—	Pass
SYS-613	The system shall hold the breakoff position automatically by adjusting drop drive amplitude.	UN-08	07	SR-700.15, RA-501, RA-502, FW-209	SSD-CSI-700; DWG-CSI-7000	T: DVP-601 TC-07	PVR-CSI-001	07.1	Pass (after ANM-0014)
SYS-614	The system shall provide a test sort function that generates side streams for adjustment without ...	UN-09	07	SR-700.16, RA-406	SSD-CSI-700; DWG-CSI-7000	D: DVP-601 TC-08	PVR-CSI-001	—	Pass
SYS-621	The system shall run a daily bead-based performance check that measures detector performance, sets ...	UN-08	10	RA-801, RA-802, AD-601	SAD-SW-CSI-011	T: DVP-601 TC-09	PVR-CSI-001	13.1	Pass
SYS-622	The system shall not allow an IVD assay to run unless the most recent performance check for the ...	UN-08, HA-08	10	RA-107 (D), RA-701, RA-803, RA-804, AD-104	SAD-SW-CSI-011	T: DVP-601 TC-10	PVR-CSI-001	08.1, 08.2, 08.3, 17.2	Pass (after ANM-0025)
SYS-623	The software configuration shall identify the installed lasers, mirrors and filters, and shall warn ...	UN-16, HA-08	10	RA-805, FW-302, AD-101, AD-102, AD-103, AD-105, AD-106	SAD-SW-CSI-011	T: DVP-601 TC-11	PVR-CSI-001	08.1	Pass
SYS-624	Performance check results shall be stored and trended so users can see changes over time.	UN-08	10	RA-806	SAD-SW-CSI-011	D: DVP-601 TC-12	PVR-CSI-001	—	Pass

ID	Requirement (abbreviated)	Source	Alloc.	Children (SR / SW)	Design output	Verification (method: test case)	Report	Risk (HS)	Result
SYS-631	The system shall allow users to define gates, sort populations and sort layouts for each collection ...	UN-02, UN-05	10	RA-306, RA-602	SAD-SW-CSI-011	D: DVP-601 TC-13	PVR-CSI-001	—	Pass
SYS-632	The system shall calculate fluorescence compensation from single-stained controls.	UN-01, UN-03	10	RA-305	SAD-SW-CSI-011	T: DVP-601 TC-14	PVR-CSI-001	—	Pass
SYS-633	The system shall show live sort counters for each sort population and the sort efficiency.	UN-02	10	RA-702, RA-704	SAD-SW-CSI-011	D: DVP-601 TC-15	PVR-CSI-001	07.3	Pass
SYS-634	The user shall be able to pause, resume and stop a sort without losing the sort record.	UN-02	10	RA-703, FW-207	SAD-SW-CSI-011	D: DVP-601 TC-16	PVR-CSI-001	15.2	Pass
SYS-635	The system shall produce a sort report giving the sort settings, counts and events for each sort.	UN-10	10	RA-307, RA-503, RA-706, RA-902, DM-101, DM-102	SAD-SW-CSI-011	D: DVP-601 TC-17	PVR-CSI-001	07.1	Pass
SRS-CSI-001 section 7									
SYS-701	The software shall require a unique user login and support at least operator, supervisor and ...	UN-10	10	AD-801, CM-101, CM-102, CM-103	SAD-SW-CSI-011	T: DVP-701 TC-01	PVR-CSI-001; SVR-SW-CSI-010	01.2, 16.2	Pass
SYS-702	The software shall keep a secure, computer-generated, time-stamped audit trail of record creation ...	UN-10, 21 CFR 11	10	FW-706, FW-808, DM-201, DM-202, DM-203, DM-204, DM-501, DM-502, DM-503	SAD-SW-CSI-011	T: DVP-701 TC-02	PVR-CSI-001; SVR-SW-CSI-010	08.4, 16.1	Pass
SYS-703	The software shall support electronic signatures for result release and assay approval.	UN-10, 21 CFR 11	10	RA-904, AD-103, AD-802, DM-301, DM-302, DM-303, DM-304	SAD-SW-CSI-011	T: DVP-701 TC-03	PVR-CSI-001; SVR-SW-CSI-010	08.4, 17.1	Pass
SYS-704	Approved IVD assay templates shall be locked so that operators cannot change instrument settings ...	UN-01, HA-08	10	RA-901, RA-902, AD-802	SAD-SW-CSI-011	T: DVP-701 TC-04	PVR-CSI-001; SVR-SW-CSI-010	08.4, 08.5, 08.6	Pass
SYS-705	The software shall export data in FCS 3.1 format and reports in PDF.	UN-10	10	RA-307, DM-401, DM-404	SAD-SW-CSI-011	T: DVP-701 TC-05	PVR-CSI-001; SVR-SW-CSI-010	17.1	Pass
SYS-706	The software shall detect and report loss of communication with the instrument without corrupting ...	UN-10	10	FW-603, FW-804, FW-805, DM-103, DM-104, DM-105	SAD-SW-CSI-011	T: DVP-701 TC-06	PVR-CSI-001; SVR-SW-CSI-010	15.2	Pass
SYS-707	The workstation shall have a locked, validated operating system image, with only the services ...	Cybersecurity	10	FW-104, FW-802, FW-803, CM-201, CM-202, CM-304	SAD-SW-CSI-011	I: DVP-701 TC-07	PVR-CSI-001; SVR-SW-CSI-010	16.1	Pass

ID	Requirement (abbreviated)	Source	Alloc.	Children (SR / SW)	Design output	Verification (method: test case)	Report	Risk (HS)	Result
SYS-708	The manufacturer shall maintain a software bill of materials and a process for assessing and ...	Cybersecurity	10	FW-104, DM-204, CM-202, CM-203	SAD-SW-CSI-011	I: DVP-701 TC-08	PVR-CSI-001; SVR-SW-CSI-010	16.1	Pass
SYS-709	User data at rest shall be protected by access control, and data exported off the workstation shall ...	UN-10, Cybersecurity	10	FW-705, DM-402, DM-403, DM-504, CM-103	SAD-SW-CSI-011	T: DVP-701 TC-09	PVR-CSI-001; SVR-SW-CSI-010	16.2	Pass
SRS-CSI-001 section 8									
SYS-801	The sorter and fluidics supply module shall connect through a keyed umbilical carrying power ...	HA-14	01	SR-100.22	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100	I: DVP-801 TC-01	PVR-CSI-001	14.1	Pass
SYS-802	The sorter shall connect to the workstation by Ethernet on a dedicated, isolated network link.	UN-10	06, 10	SR-600.16, FW-801, CM-201	SSD-CSI-600; SCH-CSI-6000; SAD-SW-CSI-011	I: DVP-801 TC-02	PVR-CSI-001	16.1	Pass
SYS-803	The system shall provide connections for the aerosol evacuation module and the temperature control ...	UN-11, UN-06	01, 09	SR-100.23, SR-900.03, RA-208	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100; SSD-CSI-900; DWG-CSI-0001	D: DVP-801 TC-03	PVR-CSI-001	—	Pass
SYS-804	Collection tube holders shall share a universal top with evacuation ports compatible with the ...	UN-06	08	SR-800.06	SSD-CSI-800; DWG-CSI-0001	I: DVP-801 TC-04	PVR-CSI-001	04.1	Pass
SYS-805	The system shall use sheath fluid of buffered saline or 1× PBS, and cleaning fluids of 10% bleach ...	UN-09	01	SR-100.24	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100	T: DVP-801 TC-05	PVR-CSI-001	—	Pass
SYS-806	The system shall support IVD reagents and QC, alignment and drop-delay beads as specified in ...	UN-01, UN-08	SYS	AD-602, AD-603	DWG-CSI-0001	T: DVP-801 TC-06	PVR-CSI-001	—	Pass
SRS-CSI-001 section 9									
SYS-901	A trained operator shall be able to complete startup, daily QC, a four-way sort and shutdown using ...	UN-09	SYS	RA-101, RA-102, RA-105, RA-1001, CM-301	DWG-CSI-0001	D: DVP-901 TC-01	PVR-CSI-001	—	Pass
SYS-902	All critical tasks identified in the use-related risk analysis shall be completed without use ...	UN-06, IEC 62366-1	SYS	CM-305	DWG-CSI-0001	T: DVP-901 TC-02	PVR-CSI-001	—	Pass (after ANM-0051)
SYS-903	Tube holders, nozzles and collection devices shall be marked with their size and keyed or shaped to ...	HA-08	02, 03, 08	SR-100.25, SR-300.05, SR-800.07	SSD-CSI-100; DWG-CSI-0001; SSD-CSI-300; DWG-CSI-3100; SSD-CSI-800	I: DVP-901 TC-03	PVR-CSI-001	—	Pass

ID	Requirement (abbreviated)	Source	Alloc.	Children (SR / SW)	Design output	Verification (method: test case)	Report	Risk (HS)	Result
SYS-904	The software shall warn the user before any action that would discard unsaved data or stop an ...	UN-10	10	RA-104	SAD-SW-CSI-011	D: DVP-901 TC-04	PVR-CSI-001	—	Pass
SYS-905	Fluid containers shall be labeled and color-coded by content, and their connectors keyed to prevent ...	HA-14	01	SR-100.26	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100	I: DVP-901 TC-05	PVR-CSI-001	14.1	Pass
SRS-CSI-001 section 10									
SYS-1001	The system shall have a design service life of at least 7 years with scheduled maintenance.	UN-12	SYS	FW-704	DWG-CSI-0001	A: DVP-1001 TC-01	PVR-CSI-001	—	Pass
SYS-1002	Mean time between failures requiring a service visit shall be at least 2,000 operating hours.	UN-12	SYS	CM-303	DWG-CSI-0001	A, T: DVP-1001 TC-02	PVR-CSI-001	—	Pass
SYS-1003	Routine user maintenance (filters, sample lines, waste caps, nozzle cleaning) shall be possible ...	UN-12	SYS	—	DWG-CSI-0001	D: DVP-1001 TC-03	PVR-CSI-001	—	Pass
SYS-1004	User maintenance intervals shall be no more frequent than weekly for filter purging and every 3 ...	UN-12	01	SR-100.27, AD-903	SSD-CSI-100; DWG-CSI-1000, DWG-CSI-1100	A: DVP-1001 TC-04	PVR-CSI-001	—	Pass
SYS-1005	Lasers, PMTs, detector arrays, PCBAs and the fluidics supply module shall be field-replaceable by a ...	UN-12	SYS	FW-301, FW-704	DWG-CSI-0001	D: DVP-1001 TC-05	PVR-CSI-001	—	Pass
SYS-1006	A trained service engineer shall be able to replace any field-replaceable unit and return the ...	UN-12	SYS	AD-902, AD-904	DWG-CSI-0001	D: DVP-1001 TC-06	PVR-CSI-001	—	Pass
SYS-1007	The system shall provide diagnostic data (pressures, laser status, voltages, error logs) to service ...	UN-12, UN-10	06, 10	SR-600.17, FW-101, FW-310, FW-312 (D), FW-313 (D), FW-701, FW-702, FW-703, FW-705, AD-201, AD-901	SSD-CSI-600; SCH-CSI-6000; SAD-SW-CSI-011	D: DVP-1001 TC-07	PVR-CSI-001	01.2	Pass
SYS-1008	Lasers and detector arrays shall be upgradable in the field according to CFG-CSI-001.	UN-16	04, 05	SR-400.14, SR-500.09, FW-313 (D), AD-101, AD-107	SSD-CSI-400; DWG-CSI-4000; SSD-CSI-500; DWG-CSI-5100, DWG-CSI-5200	D: DVP-1001 TC-08	PVR-CSI-001	—	Pass
SRS-CSI-001 section 11									
SYS-1101	Product labels shall meet ISO 15223-1 and ISO 18113-3 and shall include the UDI in human-readable ...	UN-13	SYS	—	LBL-CSI-001; OG-CSI-001	I: DVP-1101 TC-01	PVR-CSI-001	—	Pass

ID	Requirement (abbreviated)	Source	Alloc.	Children (SR / SW)	Design output	Verification (method: test case)	Report	Risk (HS)	Result
SYS-1102	The IFU (Operator's Guide) shall include intended use, limitations, warnings, installation ...	UN-13, ISO 18113-3	SYS	—	LBL-CSI-001; OG-CSI-001	I: DVP-1101 TC-02	PVR-CSI-001	01.3, 04.3, 05.1, 06.2, 07.1, 07.3, 08.5, 09.2, 17.1	Pass
SYS-1103	Laser, high-voltage, biohazard and pinch-point labels shall be placed where the hazard exists, as ...	HA-01, HA-02, HA-06	SYS	—	LBL-CSI-001; OG-CSI-001	I: DVP-1101 TC-03	PVR-CSI-001	01.1, 02.1, 06.1	Pass
SYS-1104	The system shall be supplied with a site preparation guide and a service manual.	UN-07, UN-12	SYS	—	LBL-CSI-001; OG-CSI-001	I: DVP-1101 TC-04	PVR-CSI-001	01.2, 11.1	Pass
SYS-1105	Labeling and IFU shall be available in the languages required by each market.	UN-13	SYS	CM-305	LBL-CSI-001; OG-CSI-001	I: DVP-1101 TC-05	PVR-CSI-001	—	Pass
SRS-CSI-001 section 12									
SYS-1201	The shipping configuration shall protect the sorter and the fluidics supply module so that, after ...	UN-07, UN-08	SYS	—	PKG-CSI-001; SPG-CSI-001	T: DVP-1201 TC-01	PVR-CSI-001; ISTA report	—	Pass (after ANM-0040)
SYS-1202	Shipping crates shall carry shock and tilt indicators.	UN-12	SYS	—	PKG-CSI-001; SPG-CSI-001	I: DVP-1201 TC-02	PVR-CSI-001; ISTA report	—	Pass
SYS-1203	Installation, including installation qualification, shall be completed by a service engineer in one ...	UN-07	SYS	—	PKG-CSI-001; SPG-CSI-001	D: DVP-1201 TC-03	PVR-CSI-001; ISTA report	11.1	Pass
SYS-1204	The system shall pass through a standard 90 cm (36 in) laboratory door with covers or crate removed.	UN-07	SYS	—	PKG-CSI-001; SPG-CSI-001	I: DVP-1201 TC-04	PVR-CSI-001; ISTA report	—	Pass
SRS-CSI-001 section 13									
SYS-1301	The system shall be designed and manufactured under a quality management system that meets 21 CFR ...	UN-13	SYS	AD-905	DHF-CSI-000	I: DVP-1301 TC-01	PVR-CSI-001	—	Pass
SYS-1302	The system shall meet the requirements for US marketing as a Class II IVD instrument through ...	UN-13	SYS	—	DHF-CSI-000	A: DVP-1301 TC-02	PVR-CSI-001	—	Pass
SYS-1303	The system shall meet the general safety and performance requirements of EU IVDR 2017/746 (Annex I).	UN-13	SYS	—	DHF-CSI-000	A: DVP-1301 TC-03	PVR-CSI-001	—	Pass
SYS-1304	Risk management shall be carried out according to ISO 14971:2019 and all residual risks shall be ...	UN-06, UN-13	SYS	—	DHF-CSI-000	A: DVP-1301 TC-04	PVR-CSI-001	—	Pass
SYS-1305	Software shall be developed according to IEC 62304 and IEC 82304-1.	UN-13	10	CM-401	SAD-SW-CSI-011	I: DVP-1301 TC-05	PVR-CSI-001	—	Pass

ID	Requirement (abbreviated)	Source	Alloc.	Children (SR / SW)	Design output	Verification (method: test case)	Report	Risk (HS)	Result
SYS-1306	The system shall comply with applicable restrictions on hazardous substances (EU RoHS) and with ...	UN-13	SYS	—	DHF-CSI-000	I: DVP-1301 TC-06	PVR-CSI-001	—	Pass

6. Matrix S: Subsystem Requirements and Subsystem View

Each hardware allocation in SRS-CSI-001 is refined into a subsystem requirement in the subsystem design specification (SSD). Subsystem requirements are verified at subsystem level in the subsystem verification report (SSVR-CSI-xxx) and at system level through the parent's test case in Matrix B.

6.1 Subsystem summary

Subsystem	Specification	Drawings	SYS	SR	Verification protocols	DFMEA lines (HS)
SS-01 Fluidics	SSD-CSI-100	DWG-CSI-1000, DWG-CSI-1100	19	19	DVP-301, 408, 503, 505, 601, 801, 901, 1001	Fails open (HS-09.1); Reads full when empty (HS-15.1)
SS-02 Sample Delivery	SSD-CSI-100	DWG-CSI-0001	8	8	DVP-404, 406, 408, 505, 901	Motor stall (HS-06.1)
SS-03 Flow Cell and Nozzle	SSD-CSI-300	DWG-CSI-3100	5	5	DVP-401, 405, 901	Gel void (HS-08.2)
SS-04 Excitation Optics	SSD-CSI-400	DWG-CSI-4000	15	15	DVP-301, 401, 402, 403, 502, 1001	Power drop at high base temperature (HS-08.2); Spot shift after transport (HS-08.2)
SS-05 Collection and Detection	SSD-CSI-500	DWG-CSI-5100, DWG-CSI-5200	10	10	DVP-301, 401, 402, 1001	Wrong filter in position (HS-08.1)
SS-06 Signal Processing and Control Electronics	SSD-CSI-600	SCH-CSI-6000	18	18	DVP-301, 401, 402, 403, 404, 405, 503, 801, 1001	HV drift (HS-08.2)
SS-07 Sort Mechanism	SSD-CSI-700	DWG-CSI-7000	16	16	DVP-405, 407, 503, 601	Insulation breakdown (HS-02.2); Timing error (HS-07.1)
SS-08 Sort Collection	SSD-CSI-800	DWG-CSI-0001	7	7	DVP-407, 408, 505, 801, 901	—
SS-09 Aerosol Management	SSD-CSI-900	DWG-CSI-0001	3	3	DVP-504, 801	Leak (HS-04.1)
SS-10 Workstation Software and Firmware	SRS-SW-CSI-010	SAD-SW-CSI-011	39	—	DVP-301, 402, 405, 406, 407, 408, 503, 601, 701, 801, 901, 1001, 1301	Crash during sort (HS-15.2)

SS-01 and SS-02 share SSD-CSI-100; SR counts are per subsystem. SS-10 is refined in SRS-SW-CSI-010 (Matrix C).

6.2 Subsystem requirements to parent and verification

Subsystem requirement	Parent	Parent requirement (abbreviated)	Subsystem verification	System test case
SSD-CSI-100: SS-01 Fluidics, SS-02 Sample Delivery				
SR-100.01	SYS-303	The fluidics supply module shall not exceed 66 cm height, 81 cm width ...	SSVR-CSI-100	DVP-301 TC-03
SR-100.02	SYS-304	The fluidics supply module shall be mobile on lockable casters with ...	SSVR-CSI-100	DVP-301 TC-04
SR-100.03	SYS-306	The umbilical between sorter and fluidics supply module shall allow ...	SSVR-CSI-100	DVP-301 TC-06

Subsystem requirement	Parent	Parent requirement (abbreviated)	Subsystem verification	System test case
SR-100.04	SYS-314	The fluidics supply module shall be powered and switched only from ...	SSVR-CSI-100	DVP-301 TC-10
SR-100.05	SYS-325	The system shall require no special room facilities, no external ...	SSVR-CSI-100	DVP-301 TC-17
SR-100.06	SYS-326	The system shall accept optional facility air at 80 to 100 PSI, dry ...	SSVR-CSI-100	DVP-301 TC-18
SR-100.07	SYS-453	Sample flow rate shall be user-selectable over at least 10 to 80 ...	SSVR-CSI-100	DVP-404 TC-03
SR-100.08	SYS-471	Sample carryover from one tube to the next shall not exceed 0.5% ...	SSVR-CSI-100	DVP-406 TC-01
SR-100.09	SYS-472	The system shall flush the inside and outside of the sample line with ...	SSVR-CSI-100	DVP-406 TC-02
SR-100.10	SYS-491	Sheath pressure shall be software-controlled from 5 to 75 PSI.	SSVR-CSI-100	DVP-408 TC-01
SR-100.11	SYS-492	The fluidics supply module shall hold at least 10 L of sheath, 10 L ...	SSVR-CSI-100	DVP-408 TC-02
SR-100.12	SYS-493	The system shall warn the user before sheath runs out or waste is ...	SSVR-CSI-100	DVP-408 TC-03
SR-100.13	SYS-494	The sample chamber shall accept 15 mL, 12 × 75 mm and 1 mL tubes.	SSVR-CSI-100	DVP-408 TC-04
SR-100.14	SYS-495	The system shall agitate the sample tube at user-selectable speeds.	SSVR-CSI-100	DVP-408 TC-05
SR-100.15	SYS-4101	The sample chamber shall hold a user-selected temperature, reached ...	SSVR-CSI-100	DVP-408 TC-06
SR-100.16	SYS-524	The system shall fully depressurize within 10 seconds of main power ...	SSVR-CSI-100	DVP-503 TC-04
SR-100.17	SYS-561	Pressurized tanks shall be rated for at least 1.5 times the maximum ...	SSVR-CSI-100	DVP-505 TC-01
SR-100.18	SYS-562	Moving parts (loading port, plate stage, aspirator drawer) shall not ...	SSVR-CSI-100	DVP-505 TC-02
SR-100.19	SYS-601	The system shall perform an automated fluidics startup that primes ...	SSVR-CSI-100	DVP-601 TC-01
SR-100.20	SYS-602	The system shall perform an automated fluidics shutdown that leaves ...	SSVR-CSI-100	DVP-601 TC-02
SR-100.21	SYS-603	The system shall provide automated cleaning modes: flow cell clean ...	SSVR-CSI-100	DVP-601 TC-03
SR-100.22	SYS-801	The sorter and fluidics supply module shall connect through a keyed ...	SSVR-CSI-100	DVP-801 TC-01
SR-100.23	SYS-803	The system shall provide connections for the aerosol evacuation ...	SSVR-CSI-100	DVP-801 TC-03
SR-100.24	SYS-805	The system shall use sheath fluid of buffered saline or 1× PBS, and ...	SSVR-CSI-100	DVP-801 TC-05
SR-100.25	SYS-903	Tube holders, nozzles and collection devices shall be marked with ...	SSVR-CSI-100	DVP-901 TC-03
SR-100.26	SYS-905	Fluid containers shall be labeled and color-coded by content, and ...	SSVR-CSI-100	DVP-901 TC-05

Subsystem requirement	Parent	Parent requirement (abbreviated)	Subsystem verification	System test case
SR-100.27	SYS-1004	User maintenance intervals shall be no more frequent than weekly for ...	SSVR-CSI-100	DVP-1001 TC-04
SSD-CSI-300: SS-03 Flow Cell and Nozzle				
SR-300.01	SYS-401	The coefficient of variation (CV) of the full G0/G1 peak for ...	SSVR-CSI-300	DVP-401 TC-01
SR-300.02	SYS-402	The CV of the full G0/G1 peak for Hoechst-stained chicken erythrocyte ...	SSVR-CSI-300	DVP-401 TC-02
SR-300.03	SYS-467	The system shall provide nozzles of 70, 85 and 100 µm, with a 130 µm ...	SSVR-CSI-300	DVP-405 TC-07
SR-300.04	SYS-468	Changing the nozzle shall not require optical realignment.	SSVR-CSI-300	DVP-405 TC-08
SR-300.05	SYS-903	Tube holders, nozzles and collection devices shall be marked with ...	SSVR-CSI-300	DVP-901 TC-03
SSD-CSI-400: SS-04 Excitation Optics				
SR-400.01	SYS-401	The coefficient of variation (CV) of the full G0/G1 peak for ...	SSVR-CSI-400	DVP-401 TC-01
SR-400.02	SYS-402	The CV of the full G0/G1 peak for Hoechst-stained chicken erythrocyte ...	SSVR-CSI-400	DVP-401 TC-02
SR-400.03	SYS-421	Forward and side scatter shall resolve 0.5 µm beads from instrument ...	SSVR-CSI-400	DVP-402 TC-01
SR-400.04	SYS-422	Scatter sensitivity shall allow separation of fixed platelets from ...	SSVR-CSI-400	DVP-402 TC-02
SR-400.05	SYS-431	The system shall support up to six installed lasers, with up to four ...	SSVR-CSI-400	DVP-403 TC-01
SR-400.06	SYS-432	The 488 nm and 633 nm lasers shall be standard in every ...	SSVR-CSI-400	DVP-403 TC-02
SR-400.07	SYS-433	Minimum laser power out of the laser head shall be: 375 nm > 7 mW ...	SSVR-CSI-400	DVP-403 TC-03
SR-400.08	SYS-434	Elliptical beams at the interrogation point shall measure 9 ± 3 µm in ...	SSVR-CSI-400	DVP-403 TC-04
SR-400.09	SYS-435	Each active laser shall be focused on a spatially separated beam ...	SSVR-CSI-400	DVP-403 TC-05
SR-400.10	SYS-436	Optical alignment shall be fixed at the factory and shall require no ...	SSVR-CSI-400	DVP-403 TC-06
SR-400.11	SYS-437	The system shall reach full performance within 30 minutes of laser ...	SSVR-CSI-400	DVP-403 TC-07
SR-400.12	SYS-438	Lasers shall be air-cooled and shall need no external cooling or ...	SSVR-CSI-400	DVP-403 TC-08
SR-400.13	SYS-512	Opening the flow cell access door shall block all laser beams by a ...	SSVR-CSI-400	DVP-502 TC-02
SR-400.14	SYS-1008	Lasers and detector arrays shall be upgradable in the field according ...	SSVR-CSI-400	DVP-1001 TC-08
SR-400.15	SYS-327 (D)	Every temperature-sensitive component (lasers, PMTs, cameras ...	SSVR-CSI-400	DVP-301 TC-20; DVP-403 TC-09
SSD-CSI-500: SS-05 Collection and Detection				

Subsystem requirement	Parent	Parent requirement (abbreviated)	Subsystem verification	System test case
SR-500.01	SYS-401	The coefficient of variation (CV) of the full G0/G1 peak for ...	SSVR-CSI-500	DVP-401 TC-01
SR-500.02	SYS-402	The CV of the full G0/G1 peak for Hoechst-stained chicken erythrocyte ...	SSVR-CSI-500	DVP-401 TC-02
SR-500.03	SYS-411	The doublet-to-singlet ratio for chicken erythrocyte nuclei shall be ...	SSVR-CSI-500	DVP-401 TC-03
SR-500.04	SYS-421	Forward and side scatter shall resolve 0.5 μ m beads from instrument ...	SSVR-CSI-500	DVP-402 TC-01
SR-500.05	SYS-422	Scatter sensitivity shall allow separation of fixed platelets from ...	SSVR-CSI-500	DVP-402 TC-02
SR-500.06	SYS-423	The forward scatter path shall accept interchangeable neutral density ...	SSVR-CSI-500	DVP-402 TC-03
SR-500.07	SYS-441	The system shall measure forward scatter, side scatter and at least ...	SSVR-CSI-500	DVP-402 TC-04
SR-500.08	SYS-442	Each fluorescence detector shall have a user-replaceable bandpass ...	SSVR-CSI-500	DVP-402 TC-05
SR-500.09	SYS-1008	Lasers and detector arrays shall be upgradable in the field according ...	SSVR-CSI-500	DVP-1001 TC-08
SR-500.10	SYS-327 (D)	Every temperature-sensitive component (lasers, PMTs, cameras ...	SSVR-CSI-500	DVP-301 TC-20; DVP-403 TC-09
SSD-CSI-600: SS-06 Signal Processing and Control Electronics				
SR-600.01	SYS-311	The system shall operate from 100, 115 or 230 VAC ($\pm 10\%$), 50 or 60 ...	SSVR-CSI-600	DVP-301 TC-07
SR-600.02	SYS-314	The fluidics supply module shall be powered and switched only from ...	SSVR-CSI-600	DVP-301 TC-10
SR-600.03	SYS-315	The system shall be protected by a resettable circuit breaker.	SSVR-CSI-600	DVP-301 TC-11
SR-600.04	SYS-411	The doublet-to-singlet ratio for chicken erythrocyte nuclei shall be ...	SSVR-CSI-600	DVP-401 TC-03
SR-600.05	SYS-435	Each active laser shall be focused on a spatially separated beam ...	SSVR-CSI-600	DVP-403 TC-05
SR-600.06	SYS-443	The system shall report height, area and width for every detected ...	SSVR-CSI-600	DVP-402 TC-06
SR-600.07	SYS-444	Signals from each beam spot shall be time-aligned to the same event ...	SSVR-CSI-600	DVP-402 TC-07
SR-600.08	SYS-445	Data shall be digitized with a linear dynamic range of at least 18 ...	SSVR-CSI-600	DVP-402 TC-08
SR-600.09	SYS-451	The system shall acquire at least 70,000 events per second with 8 ...	SSVR-CSI-600	DVP-404 TC-01
SR-600.10	SYS-452	The system shall acquire more than 100,000 events per second when ...	SSVR-CSI-600	DVP-404 TC-02
SR-600.11	SYS-461	For a four-way sort at 70 PSI, 90 kHz drop drive and an average ...	SSVR-CSI-600	DVP-405 TC-01
SR-600.12	SYS-464	The system shall resolve particle position within the drop to 1/32 of ...	SSVR-CSI-600	DVP-405 TC-04

Subsystem requirement	Parent	Parent requirement (abbreviated)	Subsystem verification	System test case
SR-600.13	SYS-531	The system shall provide an illuminated emergency stop button next to ...	SSVR-CSI-600	DVP-503 TC-05
SR-600.14	SYS-532	Pressing the emergency stop shall, within 2 seconds: unload the ...	SSVR-CSI-600	DVP-503 TC-06
SR-600.15	SYS-533	The emergency stop button shall light while the sample loading port ...	SSVR-CSI-600	DVP-503 TC-07
SR-600.16	SYS-802	The sorter shall connect to the workstation by Ethernet on a ...	SSVR-CSI-600	DVP-801 TC-02
SR-600.17	SYS-1007	The system shall provide diagnostic data (pressures, laser status ...	SSVR-CSI-600	DVP-1001 TC-07
SR-600.18	SYS-327 (D)	Every temperature-sensitive component (lasers, PMTs, cameras ...	SSVR-CSI-600	DVP-301 TC-20; DVP-403 TC-09
SSD-CSI-700: SS-07 Sort Mechanism				
SR-700.01	SYS-461	For a four-way sort at 70 PSI, 90 kHz drop drive and an average ...	SSVR-CSI-700	DVP-405 TC-01
SR-700.02	SYS-462	The system shall sort into up to four populations at the same time.	SSVR-CSI-700	DVP-405 TC-02
SR-700.03	SYS-463	Drop drive frequency shall be adjustable from 1 to 100 kHz or wider.	SSVR-CSI-700	DVP-405 TC-03
SR-700.04	SYS-465	The system shall automatically determine drop delay using fluorescent ...	SSVR-CSI-700	DVP-405 TC-05
SR-700.05	SYS-467	The system shall provide nozzles of 70, 85 and 100 µm, with a 130 µm ...	SSVR-CSI-700	DVP-405 TC-07
SR-700.06	SYS-483	With the plate deposition option, the system shall deposit single ...	SSVR-CSI-700	DVP-407 TC-03
SR-700.07	SYS-521	The sort collection chamber door shall not open unless the sort block ...	SSVR-CSI-700	DVP-503 TC-01
SR-700.08	SYS-522	An illuminated warning light and a software indicator shall show ...	SSVR-CSI-700	DVP-503 TC-02
SR-700.09	SYS-523	Deflection plate voltage shall not exceed 12,000 V between plates and ...	SSVR-CSI-700	DVP-503 TC-03
SR-700.10	SYS-541	The system shall monitor the drop breakoff continuously during ...	SSVR-CSI-700	DVP-503 TC-08
SR-700.11	SYS-542	If the breakoff moves outside its tolerance, the system shall pause ...	SSVR-CSI-700	DVP-503 TC-09
SR-700.12	SYS-543	If a nozzle clog is detected, the system shall, within 2 seconds ...	SSVR-CSI-700	DVP-503 TC-10
SR-700.13	SYS-611	Selecting a nozzle size in software shall load a verified default ...	SSVR-CSI-700	DVP-601 TC-05
SR-700.14	SYS-612	The system shall display a live image of the drop breakoff and of the ...	SSVR-CSI-700	DVP-601 TC-06
SR-700.15	SYS-613	The system shall hold the breakoff position automatically by ...	SSVR-CSI-700	DVP-601 TC-07
SR-700.16	SYS-614	The system shall provide a test sort function that generates side ...	SSVR-CSI-700	DVP-601 TC-08
SSD-CSI-800: SS-08 Sort Collection				

Subsystem requirement	Parent	Parent requirement (abbreviated)	Subsystem verification	System test case
SR-800.01	SYS-481	The system shall provide two-way sorting into 12 × 75 mm and 15 mL ...	SSVR-CSI-800	DVP-407 TC-01
SR-800.02	SYS-482	With the plate deposition option, the system shall sort into 6-, 24- ...	SSVR-CSI-800	DVP-407 TC-02
SR-800.03	SYS-483	With the plate deposition option, the system shall deposit single ...	SSVR-CSI-800	DVP-407 TC-03
SR-800.04	SYS-4102	With the temperature control option, collection tubes shall be held ...	SSVR-CSI-800	DVP-408 TC-07
SR-800.05	SYS-562	Moving parts (loading port, plate stage, aspirator drawer) shall not ...	SSVR-CSI-800	DVP-505 TC-02
SR-800.06	SYS-804	Collection tube holders shall share a universal top with evacuation ...	SSVR-CSI-800	DVP-801 TC-04
SR-800.07	SYS-903	Tube holders, nozzles and collection devices shall be marked with ...	SSVR-CSI-800	DVP-901 TC-03
SSD-CSI-900: SS-09 Aerosol Management				
SR-900.01	SYS-551	With the sort block and collection chamber doors closed and the ...	SSVR-CSI-900	DVP-504 TC-01
SR-900.02	SYS-552	The aerosol evacuation option shall provide at least 7 CFM in normal ...	SSVR-CSI-900	DVP-504 TC-02
SR-900.03	SYS-803	The system shall provide connections for the aerosol evacuation ...	SSVR-CSI-900	DVP-801 TC-03

7. Matrix C: Software Requirements to Design and Tests

One row per software requirement in SRS-SW-CSI-010 Rev F, grouped by software item family. Design gives the software item, the SAD section that decomposes it into units and the detailed design specification. Tests are at the level stated in the SRS-SW, and every requirement also has a software system test case. Results from SVR-SW-CSI-010 and Addendum 1 (release 3.0.2/3.0.3).

ID	Requirement (abbreviated)	Parent	Item	Units / design	Level	Test cases	Risk (HS)	Result
Part A: Research Application								
RA-101	The application shall provide a workspace with experiment browser, plots, acquisition ...	SYS-901	SI-01	SAD §6.2 (5 units); SDS-SW-CSI-012	S	STS-A-001	—	Pass
RA-102	The application shall guide the operator through startup, daily QC, sorting and shutdown with ...	SYS-901	SI-01	SAD §6.2 (5 units); SDS-SW-CSI-012	S	STS-A-002	—	Pass
RA-103	The application shall show fluid levels, system pressure, laser status, HV status, active ...	SYS-493, SYS-522	SI-01	SAD §6.2 (5 units); SDS-SW-CSI-012	S	STS-A-003	—	Pass
RA-104	The application shall ask for confirmation before any action that discards unsaved data or ...	SYS-904	SI-01	SAD §6.2 (5 units); SDS-SW-CSI-012	S	STS-A-004	—	Pass
RA-105	The application shall show progress and remaining time for each automated procedure.	SYS-901	SI-01	SAD §6.2 (5 units); SDS-SW-CSI-012	S	STS-A-005	—	Pass
RA-107 (D)	The application shall show laser base temperatures in the status panel and the performance ...	SYS-327, SYS-622, HA-08	SI-01, SI-05	SAD §6.2 (9 units); SDS-SW-CSI-012	S	STS-A-062	08.2, 17.2	Pass (SVR-SW Add. 1)
RA-201	The application shall provide an automated fluidics startup that pressurizes the system ...	SYS-601	SI-04	SAD §6.2 (4 units); SDS-SW-CSI-012	S	STS-A-006	—	Pass
RA-202	The application shall provide an automated fluidics shutdown that fills the flow path with 70% ...	SYS-602	SI-04	SAD §6.2 (4 units); SDS-SW-CSI-012	S	STS-A-007	—	Pass
RA-203	The application shall provide cleaning modes: flow cell clean, sample line backflush, prime ...	SYS-603	SI-04	SAD §6.2 (4 units); SDS-SW-CSI-012	S	STS-A-008	04.4	Pass
RA-204	The application shall not start fluidics startup unless a nozzle is detected as installed and ...	HA-09	SI-04, SI-10	SAD §6.2, §7.2 (14 units); SDS-SW-CSI-012	I	ITS §4, ITS §6, STS-A-009	—	Pass
RA-205	On Load, the application shall raise the loading port, pressurize the sample chamber and start ...	SYS-472, SYS-494	SI-04, SI-10	SAD §6.2, §7.2 (14 units); SDS-SW-CSI-012	I	ITS §4, ITS §6, STS-A-010	04.3	Pass (after ANM-0019)
RA-206	The application shall offer flow rate settings 1 to 11 (about 10 to 80 µL/min), sample ...	SYS-453, SYS-495, SYS-4101	SI-04	SAD §6.2 (4 units); SDS-SW-CSI-012	S	STS-A-011	—	Pass
RA-207	The application shall warn when sheath falls below 20% or waste rises above 80%, and shall ...	SYS-493	SI-04	SAD §6.2 (4 units); SDS-SW-CSI-012	S	STS-A-012	15.1	Pass
RA-208	The application shall detect the aerosol evacuation module, temperature control option and ...	SYS-803, SYS-551	SI-04	SAD §6.2 (4 units); SDS-SW-CSI-012	S	STS-A-013	—	Pass
RA-301	The application shall let the user set detector voltages, FSC gain, threshold parameter and ...	SYS-443	SI-02	SAD §6.2 (5 units); SDS-SW-CSI-012	S	STS-A-014	—	Pass
RA-302	The application shall display data on a linear scale from 2.6 to 262,144 or on a four- or ...	SYS-445	SI-02	SAD §6.2 (5 units); SDS-SW-CSI-012	U	UTS-010-02, STS-A-015	—	Pass

ID	Requirement (abbreviated)	Parent	Item	Units / design	Level	Test cases	Risk (HS)	Result
RA-303	The application shall acquire and display at least 70,000 events per second with 8 parameters ...	SYS-451, SYS-452	SI-02, SI-41	SAD §6.2, §6.5 (10 units); SDS-SW-CSI-012	S	STS-A-016	—	Pass
RA-304	The application shall report events lost to electronic abort or overflow and warn the user ...	SYS-451	SI-02	SAD §6.2 (5 units); SDS-SW-CSI-012	I	ITS §2, STS-A-017	08.6	Pass
RA-305	The application shall calculate a compensation matrix from single-stained controls and apply ...	SYS-632	SI-02	SAD §6.2 (5 units); SDS-SW-CSI-012	U	UTS-010-02, STS-A-018	—	Pass
RA-306	The application shall let users define polygon, rectangle, interval and quadrant gates ...	SYS-631	SI-02	SAD §6.2 (5 units); SDS-SW-CSI-012	S	STS-A-019	—	Pass
RA-307	The application shall store acquired data with all instrument settings, configuration version ...	SYS-635, SYS-705	SI-02, SI-31	SAD §6.2, §6.4 (10 units); SDS-SW-CSI-012	I	ITS §2, ITS §16, STS-A-020	—	Pass
RA-401	The application shall load the controlled default sort setup for the selected nozzle (70, 85 ...	SYS-611	SI-03	SAD §6.2 (6 units); SDS-SW-CSI-012	S	STS-A-021	—	Pass
RA-402	Changing the sort setup shall require confirmation of the installed nozzle size.	HA-08	SI-03	SAD §6.2 (6 units); SDS-SW-CSI-012	S	STS-A-022	07.2	Pass
RA-403	The application shall restore the last setup at startup with stream, breakoff control, plate ...	SYS-611, HA-02	SI-03	SAD §6.2 (6 units); SDS-SW-CSI-012	I	ITS §3, STS-A-023	—	Pass
RA-404	The application shall display the breakoff and side-stream images at least 10 times per second.	SYS-612	SI-03, SI-13	SAD §6.2, §7.2 (10 units); SDS-SW-CSI-012	I	ITS §3, ITS §9, STS-A-024	—	Pass
RA-405	The application shall let the user set drop drive frequency (1.0 to 102.0 kHz), amplitude (1.0 ...	SYS-463, SYS-611	SI-03	SAD §6.2 (6 units); SDS-SW-CSI-012	S	STS-A-025	—	Pass
RA-406	The application shall let the user turn plate voltage on and off, run a test sort, move the ...	SYS-614	SI-03	SAD §6.2 (6 units); SDS-SW-CSI-012	S	STS-A-026	—	Pass
RA-407	The application shall determine drop delay automatically with drop-delay beads by sweeping ...	SYS-465	SI-03, SI-13	SAD §6.2, §7.2 (10 units); SDS-SW-CSI-012	S	STS-A-027	07.2	Pass
RA-501	When breakoff control is On, the application shall adjust drop drive amplitude to bring Drop 1 ...	SYS-613	SI-03, SI-13	SAD §6.2, §7.2 (10 units); SDS-SW-CSI-012	I	ITS §3, ITS §9, STS-A-028	—	Pass (after ANM-0014)
RA-502	If Gap differs from its target by more than 1 pixel, the application shall adjust amplitude to ...	SYS-613	SI-03	SAD §6.2 (6 units); SDS-SW-CSI-012	I	ITS §3, STS-A-029	—	Pass
RA-503	The application shall record each sort pause and resume commanded by firmware (FW-205), with ...	SYS-635	SI-03	SAD §6.2 (6 units); SDS-SW-CSI-012	I	ITS §3, STS-A-030	—	Pass
RA-504	After a clog, the application shall show the clog recovery procedure, including running ...	SYS-554, HA-04	SI-01	SAD §6.2 (5 units); SDS-SW-CSI-012	S	STS-A-031	04.2	Pass (after ANM-0051)
RA-505	The application shall not allow the stream or plate voltage to restart after a clog until the ...	HA-04	SI-03, SI-10	SAD §6.2, §7.2 (16 units); SDS-SW-CSI-012	I	ITS §3, ITS §6, STS-A-032	04.2	Pass
RA-601	The application shall provide sort layouts for two-way 12 × 75 mm and 15 mL tubes; four-way 12 ...	SYS-481, SYS-482, SYS-485	SI-03	SAD §6.2 (6 units); SDS-SW-CSI-012	S	STS-A-033	—	Pass
RA-602	The application shall let the user assign gated populations to each sort location, set target ...	SYS-631	SI-03	SAD §6.2 (6 units); SDS-SW-CSI-012	S	STS-A-034	—	Pass

ID	Requirement (abbreviated)	Parent	Item	Units / design	Level	Test cases	Risk (HS)	Result
RA-603	The application shall not allow a four-way layout with a two-way device, or a plate layout ...	HA-08	SI-03	SAD §6.2 (6 units); SDS-SW-CSI-012	U	UTS-010-03, STS-A-035	—	Pass
RA-604	The application shall provide the read-only precision modes in Table 1 and any additional ...	SYS-464	SI-03	SAD §6.2 (6 units); SDS-SW-CSI-012	S	STS-A-036	—	Pass
RA-605	Yield and phase masks shall be mutually exclusive: setting one above 0 shall set the other to ...	SYS-464	SI-03	SAD §6.2 (6 units); SDS-SW-CSI-012	U	UTS-010-03, STS-A-037	—	Pass
RA-701	The application shall not start a sort unless the stream is on, breakoff control is on, plate ...	SYS-622, HA-08	SI-03, SI-05	SAD §6.2 (10 units); SDS-SW-CSI-012	S	STS-A-038	—	Pass
RA-702	The application shall display, for each sort location, sorted count, sort rate, conflict count ...	SYS-633	SI-01, SI-03	SAD §6.2 (11 units); SDS-SW-CSI-012	S	STS-A-039	—	Pass
RA-703	The user shall be able to pause, resume and stop a sort; stopping shall close the sort record ...	SYS-634	SI-03	SAD §6.2 (6 units); SDS-SW-CSI-012	I	ITS §3, STS-A-040	15.2	Pass
RA-704	The application shall stop the sort automatically when all target counts are reached.	SYS-633	SI-03	SAD §6.2 (6 units); SDS-SW-CSI-012	S	STS-A-041	—	Pass
RA-705	In index sorting, the application shall store the parameters of each deposited event and its ...	SYS-484	SI-03, SI-31	SAD §6.2, §6.4 (11 units); SDS-SW-CSI-012	I	ITS §3, ITS §16, STS-A-042	—	Pass
RA-706	At the end of each sort the application shall create a sort report with operator, date and ...	SYS-635	SI-03, SI-31	SAD §6.2, §6.4 (11 units); SDS-SW-CSI-012	S	STS-A-043	07.1	Pass
RA-801	The application shall run the bead-based performance check against the baseline released in ...	SYS-621	SI-05	SAD §6.2 (4 units); SDS-SW-CSI-012	S	STS-A-044	—	Pass
RA-802	The performance check shall set inter-laser delay and area scaling automatically and apply ...	SYS-444, SYS-621	SI-05, SI-11	SAD §6.2, §7.2 (12 units); SDS-SW-CSI-012	I	ITS §5, ITS §7, STS-A-045	—	Pass
RA-803	The application shall not allow acquisition or sorting with an IVD assay template unless a ...	SYS-622, HA-08	SI-05	SAD §6.2 (4 units); SDS-SW-CSI-012	S	STS-A-046	08.2, 08.3	Pass (after ANM-0025)
RA-804	A supervisor shall be able to record a documented QC lockout override for research use only ...	SYS-622	SI-05, SI-40	SAD §6.2, §6.5 (8 units); SDS-SW-CSI-012	S	STS-A-047	08.3	Pass
RA-805	The application shall warn the user before acquisition if the active lasers do not match the ...	SYS-623, HA-08	SI-05, SI-10	SAD §6.2, §7.2 (14 units); SDS-SW-CSI-012	I	ITS §5, ITS §6, STS-A-048	08.1	Pass
RA-806	The application shall display trends of detector voltage, robust CV and laser delay with ...	SYS-624	SI-05	SAD §6.2 (4 units); SDS-SW-CSI-012	S	STS-A-049	—	Pass
RA-901	When running an approved IVD assay template, the application shall lock instrument settings ...	SYS-704, HA-08	SI-05, SI-02	SAD §6.2 (9 units); SDS-SW-CSI-012	S	STS-A-050	08.4, 08.5, 08.6	Pass
RA-902	Results from an IVD assay template shall show the template name and version and whether QC ...	SYS-704, SYS-635	SI-05	SAD §6.2 (4 units); SDS-SW-CSI-012	S	STS-A-051	—	Pass
RA-903	The application shall mark any result acquired while a QC, configuration or fluidics warning ...	HA-08	SI-02, SI-05	SAD §6.2 (9 units); SDS-SW-CSI-012	S	STS-A-052	08.2	Pass
RA-904	The application shall send completed IVD results to the Data Management Application for review ...	SYS-703	SI-05, SI-31	SAD §6.2, §6.4 (9 units); SDS-SW-CSI-012	I	ITS §5, ITS §16, STS-A-053	—	Pass

ID	Requirement (abbreviated)	Parent	Item	Units / design	Level	Test cases	Risk (HS)	Result
RA-1001	The application shall classify alarms as high, medium and low, and show each with its cause ...	SYS-901	SI-01	SAD §6.2 (5 units); SDS-SW-CSI-012	S	STS-A-054	—	Pass
RA-1002	High alarms shall include clog, emergency stop, loss of instrument communication, sheath ...	HA-04, HA-06, HA-15	SI-01, SI-04	SAD §6.2 (9 units); SDS-SW-CSI-012	S	STS-A-055	—	Pass (after ANM-0023, ANM-0048)
RA-1003	The application shall show the HV-on indicator whenever firmware reports plate voltage enabled.	SYS-522, HA-02	SI-01	SAD §6.2 (5 units); SDS-SW-CSI-012	I	ITS §1, STS-A-056	02.1	Pass
Part B: Firmware								
FW-101	On power-up, SI-10 shall run a power-on self test of memory, clocks, board bus and safety ...	SYS-1007	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	I	ITS §6, STS-B-001	—	Pass
FW-102	Each FPGA shall load its configuration only after verifying its CRC; a failed check shall hold ...	HA-08	SI-11, SI-12	SAD §7.2 (14 units); SDS-FW-CSI-013	U	UTS-010-11, UTS-010-12, STS-B-002	—	Pass
FW-103	Firmware shall start in the safe state (Section 5.6) and leave it only on command from an ...	HA-02	SI-10, SI-15	SAD §7.2 (14 units); SDS-FW-CSI-013	I	ITS §6, ITS §11, STS-B-003	—	Pass (after ANM-0007)
FW-104	Firmware and FPGA images shall be accepted for update only if their digital signature is valid ...	SYS-707, SYS-708	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	I	ITS §6, STS-B-004	16.1	Pass
FW-201	SI-11 shall compute height, area and width for every enabled parameter of every event, and ...	SYS-443, SYS-444	SI-11	SAD §7.2 (8 units); SDS-FW-CSI-013	U	UTS-010-11, STS-B-005	—	Pass
FW-202	SI-11 shall digitize signals on an 18-bit linear scale and process at least 70,000 events per ...	SYS-445, SYS-451, SYS-452	SI-11	SAD §7.2 (8 units); SDS-FW-CSI-013	U	UTS-010-11, STS-B-006	—	Pass
FW-203	SI-11 shall resolve the position of each event within its drop to 1/32 of a drop and apply the ...	SYS-464	SI-11	SAD §7.2 (8 units); SDS-FW-CSI-013	U	UTS-010-11, STS-B-007	—	Pass
FW-204	SI-11 shall complete the sort decision for each event within 20 µs of the end of its event ...	SYS-461	SI-11	SAD §7.2 (8 units); SDS-FW-CSI-013	U	UTS-010-11, STS-B-008	—	Pass (after ANM-0003)
FW-205	During sorting, if Gap differs from its target by more than 2 pixels or Drop 1 by more than 10 ...	SYS-542, HA-07	SI-11, SI-13	SAD §7.2 (12 units); SDS-FW-CSI-013	I	ITS §7, ITS §9, STS-B-009	07.1	Pass
FW-206	Single-cell modes shall sort at most one target event per drop and count every sorted event ...	SYS-483	SI-11	SAD §7.2 (8 units); SDS-FW-CSI-013	U	UTS-010-11, STS-B-010	—	Pass
FW-207	Pause shall stop charging within 100 ms while acquisition continues; resume shall restart ...	SYS-634	SI-11	SAD §7.2 (8 units); SDS-FW-CSI-013	I	ITS §7, STS-B-011	—	Pass
FW-208	SI-10 shall move the plate stage to each well in layout order and enable sorting only when the ...	SYS-482, SYS-483	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	I	ITS §6, STS-B-012	—	Pass
FW-209	SI-13 shall capture strobed breakoff images at least 10 times per second and compute Drop 1 ...	SYS-612, SYS-613	SI-13	SAD §7.2 (4 units); SDS-FW-CSI-013	U	UTS-010-13, STS-B-013	—	Pass
FW-210	SI-13 shall detect a clog (loss or deflection of the stream, or Gap above the setup upper ...	SYS-541, HA-04	SI-13	SAD §7.2 (4 units); SDS-FW-CSI-013	U	UTS-010-13, STS-B-014	04.2	Pass
FW-301	Each instrument board (ADC, PMT high voltage, laser control, deflection HV, charge, motion and ...	SYS-1005	SI-12	SAD §7.2 (6 units); SDS-FW-CSI-013	I	ITS §8, STS-B-015	—	Pass

ID	Requirement (abbreviated)	Parent	Item	Units / design	Level	Test cases	Risk (HS)	Result
FW-302	At startup, SI-10 shall read every board identity, compare the inventory with the released ...	SYS-623, HA-08	SI-10, SI-12	SAD §7.2 (16 units); SDS-FW-CSI-013	I	ITS §6, ITS §8, STS-B-016	08.1	Pass
FW-303	SI-12 shall configure the ADC boards (sample rate, gain, offset) from the board calibration ...	SYS-443	SI-12	SAD §7.2 (6 units); SDS-FW-CSI-013	U	UTS-010-12, STS-B-017	—	Pass
FW-304	SI-12 shall set each PMT high voltage from 0 to 1,000 V as commanded, read it back, and report ...	SYS-441, HA-08	SI-12	SAD §7.2 (6 units); SDS-FW-CSI-013	U	UTS-010-12, STS-B-018	—	Pass (after ANM-0034)
FW-305	SI-12 shall switch each laser on and off, set laser power where the laser supports it, and ...	SYS-433, SYS-435	SI-12	SAD §7.2 (6 units); SDS-FW-CSI-013	U	UTS-010-12, STS-B-019	—	Pass
FW-306	SI-12 shall enforce that lasers sharing a beam spot (488/445 and 405/375) are never on at the ...	SYS-435	SI-12	SAD §7.2 (6 units); SDS-FW-CSI-013	U	UTS-010-12, STS-B-020	—	Pass
FW-307	SI-12 shall set deflection plate voltage and stream charge levels as commanded, read back ...	SYS-523, HA-02	SI-12	SAD §7.2 (6 units); SDS-FW-CSI-013	U	UTS-010-12, STS-B-021	02.2	Pass (after ANM-0066)
FW-308	SI-12 shall monitor supply rails (within ±5%), board temperatures and fan speeds, and report ...	HA-12	SI-12	SAD §7.2 (6 units); SDS-FW-CSI-013	U	UTS-010-12, STS-B-022	12.2	Pass
FW-309	SI-12 shall drive the loading port, aspirator drawer, drop-delay filter and plate stage motors ...	SYS-562	SI-12	SAD §7.2 (6 units); SDS-FW-CSI-013	U	UTS-010-12, STS-B-023	06.1	Pass
FW-310	SI-12 shall expose board status and counters through the register map defined in ...	SYS-1007	SI-12	SAD §7.2 (6 units); SDS-FW-CSI-013	I	ITS §8, STS-B-024	—	Pass
FW-311 (D)	SI-10 shall control fan duty from the hottest laser base temperature reported by SI-12, with a ...	SYS-327, HA-08	SI-10, SI-12	SAD §7.2 (16 units); SDS-FW-CSI-013	I	ITS §6, ITS §8, STS-B-053	08.2, 17.2	Pass (SVR-SW Add. 1)
FW-401	SI-14 shall hold sheath pressure within ±1% of the setpoint across 5 to 75 PSI and report ...	SYS-491	SI-14	SAD §7.2 (4 units); SDS-FW-CSI-013	U	UTS-010-14, STS-B-025	09.1	Pass
FW-402	SI-14 shall report sheath, waste, ethanol and cleaning fluid levels and supply air pressure ...	SYS-493	SI-14, SI-10	SAD §7.2 (14 units); SDS-FW-CSI-013	I	ITS §10, ITS §6, STS-B-026	15.1	Pass
FW-403	On main power off, SI-14 shall vent system pressure within 10 seconds.	SYS-524	SI-14	SAD §7.2 (4 units); SDS-FW-CSI-013	I	ITS §10, STS-B-027	09.2	Pass
FW-501	The loading port shall move only while the emergency stop is not active, and SI-10 shall light ...	SYS-533	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	I	ITS §6, STS-B-028	—	Pass
FW-502	SI-10 shall monitor door interlock, laser shutter and emergency stop status and reject stream ...	SYS-512, SYS-521	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	I	ITS §6, STS-B-029	01.1, 05.1	Pass
FW-601	On emergency stop, SI-10 shall unload the sample tube, stop the plate stage, turn off stream ...	SYS-532, HA-02, HA-06	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	I	ITS §6, STS-B-030	06.2	Pass
FW-602	On a detected clog, SI-10 shall, without waiting for the workstation, stop the stream, turn ...	SYS-543, HA-04	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	I	ITS §6, STS-B-031	04.2, 15.2	Pass (after ANM-0011)
FW-603	If communication with the workstation is lost for more than 2 seconds during sorting, SI-10 ...	SYS-706, HA-02	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	I	ITS §6, STS-B-032	—	Pass
FW-604	SI-15 shall monitor heartbeats from SI-10 and SI-11; if either is missing for more than 250 ...	HA-02	SI-15	SAD §7.2 (4 units); SDS-FW-CSI-013	U	UTS-010-15, STS-B-033	02.3	Pass (after ANM-0007)

ID	Requirement (abbreviated)	Parent	Item	Units / design	Level	Test cases	Risk (HS)	Result
FW-605	SI-10 and SI-15 shall each run a hardware watchdog; a watchdog timeout shall put the ...	HA-02	SI-10, SI-15	SAD §7.2 (14 units); SDS-FW-CSI-013	U	UTS-010-10, UTS-010-15, STS-B-034	02.3	Pass
FW-606	A board fault reported under FW-304, FW-307 or FW-308, an FPGA configuration failure or an ...	HA-02, HA-08, HA-12	SI-10, SI-12	SAD §7.2 (16 units); SDS-FW-CSI-013	I	ITS §6, ITS §8, STS-B-035	02.2, 12.2	Pass
FW-607	The safe state shall be latched: firmware shall leave it only after the cause has cleared and ...	HA-02, HA-04	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	I	ITS §6, STS-B-036	—	Pass
FW-608	SI-10 shall report HV status at least 5 times per second and report every entry into the safe ...	SYS-522	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	I	ITS §6, STS-B-037	—	Pass
FW-701	SI-10 shall keep an internal log of events, warnings, faults, safe-state entries, commands ...	SYS-1007	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	I	ITS §6, STS-B-038	—	Pass
FW-702	Each log entry shall contain a UTC time stamp with millisecond resolution synchronized to the ...	SYS-1007	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	U	UTS-010-10, STS-B-039	—	Pass
FW-703	The log shall be kept in non-volatile memory for at least 100,000 entries or 90 days ...	SYS-1007	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	I	ITS §6, STS-B-040	—	Pass (after ANM-0035)
FW-704	SI-10 shall keep usage counters for laser on-hours per laser, PMT on-hours, stream hours, sort ...	SYS-1001, SYS-1005	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	U	UTS-010-10, STS-B-041	—	Pass
FW-705	The log shall contain no patient or specimen data, and shall be readable and exportable only ...	SYS-1007, SYS-709	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	I	ITS §6, STS-B-042	—	Pass
FW-706	Log entries shall not be editable or deletable by any user; the log export shall include a ...	SYS-702	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	U	UTS-010-10, STS-B-043	—	Pass
FW-312 (D)	SI-10 shall log each laser base temperature once per minute and every fan-duty change, with ...	SYS-1007, SYS-327	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	I	ITS §6, STS-B-054	—	Pass (SVR-SW Add. 1)
FW-313 (D)	SI-10 shall report the firmware version of every purchased module containing firmware to the ...	SYS-1007, SYS-1008	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	I	ITS §6, STS-B-055	—	Pass (SVR-SW Add. 1)
FW-801	The instrument shall communicate with the workstation over a dedicated Ethernet link using the ...	SYS-802	SI-10, SI-41	SAD §7.2, §6.5 (15 units); SDS-FW-CSI-013	I	ITS §6, ITS §19, STS-B-044	—	Pass
FW-802	The instrument shall accept connections only from the paired workstation, authenticated at ...	SYS-707	SI-10, SI-41	SAD §7.2, §6.5 (15 units); SDS-FW-CSI-013	I	ITS §6, ITS §19, STS-B-045	16.1	Pass (after ANM-0030)
FW-803	Every message shall carry a sequence number and a CRC-32; messages with a bad CRC, a sequence ...	SYS-707, HA-08	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	U	UTS-010-10, STS-B-046	16.1	Pass (after ANM-0030)
FW-804	The instrument shall acknowledge every command within 100 ms with accepted, rejected (with ...	SYS-706	SI-10	SAD §7.2 (10 units); SDS-FW-CSI-013	U	UTS-010-10, STS-B-047	—	Pass
FW-805	Instrument and workstation shall exchange heartbeats every 250 ms; loss for more than 2 ...	SYS-706	SI-10, SI-41	SAD §7.2, §6.5 (15 units); SDS-FW-CSI-013	I	ITS §6, ITS §19, STS-B-048	—	Pass (after ANM-0023)
FW-806	The event data channel shall carry at least 110,000 events per second at 8 parameters without ...	SYS-451, SYS-452	SI-10, SI-11, SI-41	SAD §7.2, §6.5 (23 units); SDS-FW-CSI-013	I	ITS §6, ITS §7, ITS §19, STS-B-049	—	Pass
FW-807	Workstation and instrument shall check protocol version compatibility at connection and refuse ...	HA-08	SI-10, SI-41	SAD §7.2, §6.5 (15 units); SDS-FW-CSI-013	I	ITS §6, ITS §19, STS-B-050	—	Pass

ID	Requirement (abbreviated)	Parent	Item	Units / design	Level	Test cases	Risk (HS)	Result
FW-808	The instrument clock shall be synchronized to the workstation at connection and at least ...	SYS-702	SI-10, SI-41	SAD §7.2, §6.5 (15 units); SDS-FW-CSI-013	I	ITS §6, ITS §19, STS-B-051	—	Pass
Part C: Admin Application								
AD-101	The application shall maintain a controlled hardware configuration record for each instrument ...	SYS-623, SYS-1008	SI-20	SAD §6.3 (4 units); SDS-SW-CSI-012	S	STS-C-001	—	Pass
AD-102	The application shall compare the configuration record with the board inventory reported by ...	SYS-623, HA-08	SI-20, SI-10	SAD §6.3, §7.2 (14 units); SDS-SW-CSI-012	I	ITS §12, ITS §6, STS-C-002	—	Pass
AD-103	Each change to the configuration record shall require the administrator or service role, a ...	SYS-623, SYS-703	SI-20, SI-32	SAD §6.3, §6.4 (7 units); SDS-SW-CSI-012	S	STS-C-003	—	Pass
AD-104	After a configuration change affecting lasers, optics or detectors, the application shall ...	SYS-622, HA-08	SI-20, SI-05	SAD §6.3, §6.2 (8 units); SDS-SW-CSI-012	S	STS-C-004	08.1	Pass
AD-105	The application shall enforce the configuration rules in CFG-CSI-001, including: unique ...	SYS-623	SI-20	SAD §6.3 (4 units); SDS-SW-CSI-012	U	UTS-010-20, STS-C-005	08.1	Pass (after ANM-0026)
AD-106	The application shall generate the optical configuration (parameter names, detector positions ...	SYS-623	SI-20	SAD §6.3 (4 units); SDS-SW-CSI-012	I	ITS §12, STS-C-006	08.1	Pass
AD-107	The application shall record field upgrades of lasers and detector arrays as configuration ...	SYS-1008	SI-20	SAD §6.3 (4 units); SDS-SW-CSI-012	S	STS-C-007	—	Pass
AD-201	The application shall display, for each laser, measured power, current, temperature, on-hours ...	SYS-433, SYS-1007	SI-21	SAD §6.3 (6 units); SDS-SW-CSI-012	S	STS-C-008	—	Pass
AD-202	The application shall provide a laser power calibration that records power measured at the ...	SYS-433	SI-21	SAD §6.3 (6 units); SDS-SW-CSI-012	S	STS-C-009	—	Pass (after ANM-0016)
AD-203	The application shall flag any laser whose power at the head is below the minimum in ...	SYS-433, HA-08	SI-21, SI-05	SAD §6.3, §6.2 (10 units); SDS-SW-CSI-012	S	STS-C-010	—	Pass
AD-204	The application shall provide a warm-up verification that records laser power stability over ...	SYS-437	SI-21	SAD §6.3 (6 units); SDS-SW-CSI-012	S	STS-C-011	—	Pass
AD-205	The application shall calibrate inter-laser delay and area scaling for each laser and nozzle ...	SYS-444	SI-21	SAD §6.3 (6 units); SDS-SW-CSI-012	S	STS-C-012	—	Pass
AD-301	The application shall provide an alignment verification tool that runs alignment beads and ...	SYS-436	SI-21	SAD §6.3 (6 units); SDS-SW-CSI-012	S	STS-C-013	—	Pass
AD-302	The application shall calibrate each PMT's voltage-to-gain curve with multi-intensity beads ...	SYS-441	SI-21	SAD §6.3 (6 units); SDS-SW-CSI-012	S	STS-C-014	—	Pass
AD-303	The application shall verify detector linearity with multi-peak beads and report the deviation ...	SYS-411	SI-21	SAD §6.3 (6 units); SDS-SW-CSI-012	S	STS-C-015	—	Pass
AD-304	The application shall provide a resolution check with chicken erythrocyte nuclei that reports ...	SYS-401, SYS-402, SYS-411	SI-21	SAD §6.3 (6 units); SDS-SW-CSI-012	S	STS-C-016	—	Pass
AD-305	The application shall record the installed FSC ND filter and apply the matching FSC scaling.	SYS-423	SI-21, SI-20	SAD §6.3 (10 units); SDS-SW-CSI-012	S	STS-C-017	—	Pass

ID	Requirement (abbreviated)	Parent	Item	Units / design	Level	Test cases	Risk (HS)	Result
AD-401	The application shall calibrate each pressure sensor at two points against a calibrated ...	SYS-491	SI-21, SI-14	SAD §6.3, §7.2 (10 units); SDS-SW-CSI-012	I	ITS §13, ITS §10, STS-C-018	—	Pass
AD-402	The application shall calibrate sample flow rate by recording the sample volume consumed ...	SYS-453	SI-21	SAD §6.3 (6 units); SDS-SW-CSI-012	S	STS-C-019	—	Pass
AD-403	The application shall calibrate each level sensor at empty and full.	SYS-493	SI-21, SI-14	SAD §6.3, §7.2 (10 units); SDS-SW-CSI-012	I	ITS §13, ITS §10, STS-C-020	—	Pass
AD-404	The application shall run a pressure decay leak test of the sheath and sample paths and report ...	SYS-491	SI-21	SAD §6.3 (6 units); SDS-SW-CSI-012	S	STS-C-021	—	Pass
AD-405	The application shall calibrate the sample chamber temperature sensor against a reference ...	SYS-4101	SI-21	SAD §6.3 (6 units); SDS-SW-CSI-012	S	STS-C-022	—	Pass
AD-501	The application shall calibrate breakoff and side-stream camera scale and focus with the ...	SYS-612	SI-21, SI-13	SAD §6.3, §7.2 (10 units); SDS-SW-CSI-012	I	ITS §13, ITS §9, STS-C-023	—	Pass
AD-502	The application shall verify drop drive frequency and plate voltage against readback and ...	SYS-463, SYS-523	SI-21, SI-12	SAD §6.3, §7.2 (12 units); SDS-SW-CSI-012	I	ITS §13, ITS §8, STS-C-024	—	Pass
AD-503	The application shall measure stream position in the waste aspirator for each nozzle and guide ...	SYS-468	SI-21	SAD §6.3 (6 units); SDS-SW-CSI-012	S	STS-C-025	—	Pass
AD-504	The application shall calibrate the plate stage home position and the first and last well of ...	SYS-483	SI-21, SI-12	SAD §6.3, §7.2 (12 units); SDS-SW-CSI-012	S	STS-C-026	—	Pass (after ANM-0037)
AD-505	The application shall verify collection temperature control against a reference thermometer ...	SYS-4102	SI-21	SAD §6.3 (6 units); SDS-SW-CSI-012	S	STS-C-027	—	Pass
AD-601	The application shall create a performance check baseline for each optical configuration at ...	SYS-621	SI-22, SI-21	SAD §6.3 (11 units); SDS-SW-CSI-012	S	STS-C-028	—	Pass
AD-602	The application shall import bead lot files with target values and limits, and reject files ...	SYS-806, HA-08	SI-22	SAD §6.3 (5 units); SDS-SW-CSI-012	U	UTS-010-22, STS-C-029	—	Pass
AD-603	The application shall manage drop-delay bead and alignment bead lots with expiry dates.	SYS-806	SI-22	SAD §6.3 (5 units); SDS-SW-CSI-012	S	STS-C-030	—	Pass
AD-701	The application shall hold the default sort setups for each nozzle as controlled records ...	SYS-611	SI-22, SI-32	SAD §6.3, §6.4 (8 units); SDS-SW-CSI-012	S	STS-C-031	—	Pass
AD-702	The application shall let an administrator define custom collection devices by rows, columns ...	SYS-485	SI-22	SAD §6.3 (5 units); SDS-SW-CSI-012	S	STS-C-032	—	Pass
AD-703	The application shall let an administrator create, name and delete additional precision modes ...	SYS-464	SI-22	SAD §6.3 (5 units); SDS-SW-CSI-012	S	STS-C-033	—	Pass
AD-801	The application shall let an administrator create, disable and assign roles to user accounts ...	SYS-701	SI-22, SI-40	SAD §6.3, §6.5 (9 units); SDS-SW-CSI-012	S	STS-C-034	—	Pass
AD-802	The application shall let a supervisor approve an IVD assay template by electronic signature ...	SYS-704, SYS-703	SI-22, SI-32	SAD §6.3, §6.4 (8 units); SDS-SW-CSI-012	S	STS-C-035	08.4	Pass
AD-901	The application shall display firmware diagnostics (pressures, laser status, voltages ...	SYS-1007	SI-21	SAD §6.3 (6 units); SDS-SW-CSI-012	S	STS-C-036	01.2	Pass
AD-902	Each calibration shall be recorded with as-found and as-left values, pass or fail, reference ...	SYS-1006	SI-21, SI-31	SAD §6.3, §6.4 (11 units); SDS-SW-CSI-012	S	STS-C-037	—	Pass

ID	Requirement (abbreviated)	Parent	Item	Units / design	Level	Test cases	Risk (HS)	Result
AD-903	The application shall warn when a calibration is due within 30 days and block IVD templates ...	SYS-1004, HA-08	SI-21, SI-05	SAD §6.3, §6.2 (10 units); SDS-SW-CSI-012	S	STS-C-038	—	Pass
AD-904	The application shall produce a calibration certificate report for each instrument.	SYS-1006	SI-21	SAD §6.3 (6 units); SDS-SW-CSI-012	S	STS-C-039	—	Pass
AD-905	Calibration tools used in manufacturing shall produce records in the format required by ...	SYS-1301	SI-21	SAD §6.3 (6 units); SDS-SW-CSI-012	S	STS-C-040	—	Pass
Part D: Data Management Application								
DM-101	The application shall let users browse and search experiments, tubes, sort records, QC results ...	SYS-635	SI-30	SAD §6.4 (5 units); SDS-SW-CSI-012	S	STS-D-001	—	Pass
DM-102	The application shall display stored data, plots and statistics without an instrument ...	SYS-635	SI-30	SAD §6.4 (5 units); SDS-SW-CSI-012	S	STS-D-002	—	Pass
DM-103	SI-31 shall use transactions so each record is written completely or not at all, and shall ...	SYS-706	SI-31	SAD §6.4 (5 units); SDS-SW-CSI-012	U	UTS-010-31, STS-D-003	—	Pass
DM-104	On workstation power loss or crash, data recorded up to the last completed second, sort ...	SYS-706	SI-31	SAD §6.4 (5 units); SDS-SW-CSI-012	S	STS-D-004	15.2	Pass
DM-105	The application shall warn the administrator when database storage exceeds 80% of its ...	SYS-706	SI-31, SI-30	SAD §6.4 (10 units); SDS-SW-CSI-012	S	STS-D-005	—	Pass
DM-201	SI-32 shall record every creation, modification and deletion of experiments, data, templates ...	SYS-702	SI-32	SAD §6.4 (3 units); SDS-SW-CSI-012	I	ITS §17, STS-D-006	08.4, 16.1	Pass (after ANM-0028)
DM-202	Audit trail entries shall be generated by the software within 1 second of the action ...	SYS-702	SI-32	SAD §6.4 (3 units); SDS-SW-CSI-012	U	UTS-010-32, STS-D-007	—	Pass
DM-203	Supervisors shall be able to view, filter and export the audit trail.	SYS-702	SI-30	SAD §6.4 (5 units); SDS-SW-CSI-012	S	STS-D-008	—	Pass
DM-204	Security events (logins, failed logins, account changes, update attempts) shall be recorded in ...	SYS-702, SYS-708	SI-32, SI-40	SAD §6.4, §6.5 (7 units); SDS-SW-CSI-012	S	STS-D-009	—	Pass
DM-301	IVD results shall have status Draft, Reviewed or Released; only released results can be ...	SYS-703	SI-30	SAD §6.4 (5 units); SDS-SW-CSI-012	S	STS-D-010	17.1	Pass
DM-302	Electronic signatures shall require re-entry of the password and record the signer's name ...	SYS-703	SI-32	SAD §6.4 (3 units); SDS-SW-CSI-012	S	STS-D-011	—	Pass
DM-303	Signatures shall be linked to their records so that they cannot be copied or transferred to ...	SYS-703	SI-32	SAD §6.4 (3 units); SDS-SW-CSI-012	U	UTS-010-32, STS-D-012	—	Pass
DM-304	Changing a released record shall create a new version with reason and signature; the released ...	SYS-703	SI-30, SI-32	SAD §6.4 (8 units); SDS-SW-CSI-012	S	STS-D-013	—	Pass
DM-401	The application shall export data in FCS 3.1 format with all keywords needed to reproduce the ...	SYS-705	SI-30	SAD §6.4 (5 units); SDS-SW-CSI-012	U	UTS-010-30, STS-D-014	—	Pass (after ANM-0045)
DM-402	Exported files shall include a checksum so that later changes can be detected.	SYS-709	SI-30	SAD §6.4 (5 units); SDS-SW-CSI-012	U	UTS-010-30, STS-D-015	—	Pass
DM-403	The application shall offer an export option that removes patient identifiers.	SYS-709	SI-30	SAD §6.4 (5 units); SDS-SW-CSI-012	S	STS-D-016	16.2	Pass

ID	Requirement (abbreviated)	Parent	Item	Units / design	Level	Test cases	Risk (HS)	Result
DM-404	The application shall export released IVD results to a laboratory information system in a ...	SYS-705	SI-30	SAD §6.4 (5 units); SDS-SW-CSI-012	S	STS-D-017	17.1	Pass (after ANM-0058)
DM-501	The application shall provide scheduled and manual backup of the database and data files, and ...	SYS-702	SI-30	SAD §6.4 (5 units); SDS-SW-CSI-012	S	STS-D-018	—	Pass
DM-502	The application shall archive records to external storage with integrity checks, and retrieve ...	SYS-702	SI-30	SAD §6.4 (5 units); SDS-SW-CSI-012	S	STS-D-019	—	Pass
DM-503	Records shall be retained for the periods in Section 9; deletion shall be possible only for an ...	SYS-702	SI-30, SI-32	SAD §6.4 (8 units); SDS-SW-CSI-012	S	STS-D-020	—	Pass
DM-504	User data at rest shall be encrypted and accessible only through the applications.	SYS-709	SI-31	SAD §6.4 (5 units); SDS-SW-CSI-012	I	ITS §16, STS-D-021	16.2	Pass
Common (security and communication)								
CM-101	All applications shall require login with a unique user ID and password through SI-40 before ...	SYS-701	SI-40	SAD §6.5 (4 units); SDS-SW-CSI-012	S	STS-E-001	16.2	Pass
CM-102	SI-40 shall provide operator, supervisor, administrator and service roles with the permissions ...	SYS-701	SI-40	SAD §6.5 (4 units); SDS-SW-CSI-012	S	STS-E-002	01.2	Pass
CM-103	Passwords shall have configurable length, complexity and expiry; accounts shall lock after a ...	SYS-701, SYS-709	SI-40	SAD §6.5 (4 units); SDS-SW-CSI-012	U	UTS-010-40, STS-E-003	—	Pass (after ANM-0042)
CM-201	The workstation image shall enable only the services and ports needed for operation; the ...	SYS-707, SYS-802	SI-41	SAD §6.5 (5 units); SDS-SW-CSI-012	S	STS-E-004	16.1	Pass
CM-202	Application updates shall be accepted only if their digital signature is valid.	SYS-707, SYS-708	SI-30	SAD §6.4 (5 units); SDS-SW-CSI-012	I	ITS §15, STS-D-022	—	Pass
CM-203	A software bill of materials covering applications and firmware shall be generated for every ...	SYS-708	SI-10, SI-30	SAD §7.2, §6.4 (15 units); SDS-FW-CSI-013	S	STS-B-052	—	Pass
CM-301	Applications shall update plots at least twice per second at the maximum event rate and ...	SYS-901	SI-01	SAD §6.2 (5 units); SDS-SW-CSI-012	S	STS-A-057	—	Pass
CM-302	Each application shall start and, where needed, connect to the instrument within 60 seconds.	SYS-604	SI-01, SI-41	SAD §6.2, §6.5 (10 units); SDS-SW-CSI-012	S	STS-A-058	—	Pass
CM-303	The software system shall run 24 hours of continuous acquisition at the maximum event rate ...	SYS-1002	SI-02, SI-10	SAD §6.2, §7.2 (15 units); SDS-SW-CSI-012	S	STS-A-059	—	Pass (after ANM-0044)
CM-304	Applications shall run on the validated workstation configuration in the release notes and ...	SYS-707	SI-30	SAD §6.4 (5 units); SDS-SW-CSI-012	S	STS-D-023	—	Pass
CM-305	User interface text shall be stored in resource files to support translation, and the user ...	SYS-1105, SYS-902	SI-01	SAD §6.2 (5 units); SDS-SW-CSI-012	I	ITS §1, STS-A-060	—	Pass; deferred: ANM-0053
CM-401	All software items shall be developed, verified, released and maintained under SDP-CSI-010 to ...	SYS-1305	SI-01 to SI-41 (all items)	SAD §6.2, §6.5 (10 units); SDS-SW-CSI-012	S	STS-A-061	—	Pass

8. Matrix D: Risk Controls to Requirements and Verification

One row per risk control requirement for each hazardous situation in RMF-CSI-001 Rev D. Control type follows ISO 14971 clause 7.1 (ID inherent safety by design, PM protective measure, IS information for safety). Implementation is verified by the requirement's test case; effectiveness evidence is the RMF record. Residual risk is from RMF Table 6-2.

HS	Harm (S)	Type	Risk control measure	Req.	Verification of implementation	Effectiveness evidence	Result	Residual
HA-01: Laser radiation (Class 3B, 375 to 633 nm)								
HS-01.1	H-01 (S4)	ID	Enclosed beam path; Class 1 in all user-accessible conditions	SYS-511	DVP-502 TC-01; SVR-CSI-001	Accessible emission below Class 1 AEL with door open (all builds); shutter fault injection	Pass	S4 P1 A
HS-01.1	H-01 (S4)	PM	Hardware shutter blocks all beams when door opens; firmware rejects conflicting ...	SYS-512	DVP-502 TC-02; SVR-CSI-001	Accessible emission below Class 1 AEL with door open (all builds); shutter fault injection	Pass	S4 P1 A
HS-01.1	H-01 (S4)	PM	Hardware shutter blocks all beams when door opens; firmware rejects conflicting ...	FW-502	I: ITS §6, STS-B-029; SVR-SW-CSI-010	Accessible emission below Class 1 AEL with door open (all builds); shutter fault injection	Pass	S4 P1 A
HS-01.1	H-01 (S4)	IS	Warning labels; IFU	SYS-513	DVP-502 TC-03; SVR-CSI-001	Accessible emission below Class 1 AEL with door open (all builds); shutter fault injection	Pass (after ANM-0032)	S4 P1 A
HS-01.1	H-01 (S4)	IS	Warning labels; IFU	SYS-1103	DVP-1101 TC-03; PVR-CSI-001	Accessible emission below Class 1 AEL with door open (all builds); shutter fault injection	Pass	S4 P1 A
HS-01.2	H-01 (S4)	ID	Factory-fixed alignment; no routine field alignment of launch plate	SYS-436	DVP-403 TC-06; PVR-CSI-001	Service validation at beta sites (VAL-CSI-10): procedure followed in 100% of observed ...	Pass	S4 P1 A
HS-01.2	H-01 (S4)	PM	Service override requires Admin Application service role and key; auto-expires ...	AD-901	S: STS-C-036; SVR-SW-CSI-010	Service validation at beta sites (VAL-CSI-10): procedure followed in 100% of observed ...	Pass	S4 P1 A
HS-01.2	H-01 (S4)	PM	Service override requires Admin Application service role and key; auto-expires ...	CM-102	S: STS-E-002; SVR-SW-CSI-010	Service validation at beta sites (VAL-CSI-10): procedure followed in 100% of observed ...	Pass	S4 P1 A
HS-01.2	H-01 (S4)	IS	Service manual laser safety procedure, eyewear by wavelength, training	SYS-1104	DVP-1101 TC-04; PVR-CSI-001	Service validation at beta sites (VAL-CSI-10): procedure followed in 100% of observed ...	Pass	S4 P1 A
HS-01.3	H-02 (S2)	ID	Camera window with laser-line filter	SYS-511	DVP-502 TC-01; SVR-CSI-001	Emission measurement at window	Pass	S2 P1 A
HS-01.3	H-02 (S2)	IS	IFU	SYS-1102	DVP-1101 TC-02; PVR-CSI-001	Emission measurement at window	Pass	S2 P1 A
HA-02: High voltage on deflection plates (up to 12 kV)								
HS-02.1	H-03 (S4)	ID	Plates recessed and insulated; no energized part accessible with door closed	SYS-523	DVP-503 TC-03; PVR-CSI-001	VAL-CSI-03 critical task; pinch and reach test with standard test finger	Pass	S4 P2 R

HS	Harm (S)	Type	Risk control measure	Req.	Verification of implementation	Effectiveness evidence	Result	Residual
HS-02.1	H-03 (S4)	PM	Red warning light and on-screen indicator; HV removed by e-stop	SYS-522	DVP-503 TC-02; PVR-CSI-001	VAL-CSI-03 critical task; pinch and reach test with standard test finger	Pass	S4 P2 R
HS-02.1	H-03 (S4)	PM	Red warning light and on-screen indicator; HV removed by e-stop	SYS-532	DVP-503 TC-06; PVR-CSI-001	VAL-CSI-03 critical task; pinch and reach test with standard test finger	Pass	S4 P2 R
HS-02.1	H-03 (S4)	PM	Red warning light and on-screen indicator; HV removed by e-stop	RA-1003	I: ITS §1, STS-A-056; SVR-SW-CSI-010	VAL-CSI-03 critical task; pinch and reach test with standard test finger	Pass	S4 P2 R
HS-02.1	H-03 (S4)	IS	HV label at sort block; IFU cleaning procedure requires HV off	SYS-1103	DVP-1101 TC-03; PVR-CSI-001	VAL-CSI-03 critical task; pinch and reach test with standard test finger	Pass	S4 P2 R
HS-02.2	H-03 (S4)	ID	Creepage and clearance per IEC 61010-1; sealed HV feedthroughs; drip channels	SYS-501	DVP-501 TC-01; SVR-CSI-001	Fluid ingress test with saline; fault injection	Pass	S4 P1 A
HS-02.2	H-03 (S4)	ID	Creepage and clearance per IEC 61010-1; sealed HV feedthroughs; drip channels	SYS-523	DVP-503 TC-03; PVR-CSI-001	Fluid ingress test with saline; fault injection	Pass	S4 P1 A
HS-02.2	H-03 (S4)	PM	HV readback fault triggers safe state; 100% hipot in production	FW-307	U: UTS-010-12, STS-B-021; SVR-SW-CSI-010	Fluid ingress test with saline; fault injection	Pass (after ANM-0066)	S4 P1 A
HS-02.2	H-03 (S4)	PM	HV readback fault triggers safe state; 100% hipot in production	FW-606	I: ITS §6, ITS §8, STS-B-035; SVR-SW-CSI-010	Fluid ingress test with saline; fault injection	Pass	S4 P1 A
HS-02.2	H-03 (S4)	PM	HV readback fault triggers safe state; 100% hipot in production	SYS-502	DVP-501 TC-02; SVR-CSI-001	Fluid ingress test with saline; fault injection	Pass	S4 P1 A
HS-02.3	H-03 (S4)	PM	Independent safety supervisor removes HV enable within 250 ms of missing ...	FW-604	U: UTS-010-15, STS-B-033; SVR-SW-CSI-010	Fault injection: SI-10 hang, SI-11 hang, bus fault (59 trials each)	Pass (after ANM-0007)	S4 P1 A
HS-02.3	H-03 (S4)	PM	Independent safety supervisor removes HV enable within 250 ms of missing ...	FW-605	U: UTS-010-10, UTS-010-15, STS-B-034; SVR-SW-CSI-010	Fault injection: SI-10 hang, SI-11 hang, bus fault (59 trials each)	Pass	S4 P1 A
HS-02.3	H-03 (S4)	PM	Hardware warning light driven from HV supply, not from software	SYS-522	DVP-503 TC-02; PVR-CSI-001	Fault injection: SI-10 hang, SI-11 hang, bus fault (59 trials each)	Pass	S4 P1 A
HA-04: Infectious aerosol from droplet sorting								
HS-04.1	H-05 (S4)	ID	Sealed sort block and collection chamber	SYS-804	DVP-801 TC-04; PVR-CSI-001	Fluorescent microsphere aerosol challenge (3 conditions × 3 runs)	Pass	S4 P1 A
HS-04.1	H-05 (S4)	PM	Aerosol evacuation module with ULPA filter, ≥ 7 CFM	SYS-551	DVP-504 TC-01; PVR-CSI-001	Fluorescent microsphere aerosol challenge (3 conditions × 3 runs)	Pass	S4 P1 A
HS-04.1	H-05 (S4)	PM	Aerosol evacuation module with ULPA filter, ≥ 7 CFM	SYS-552	DVP-504 TC-02; PVR-CSI-001	Fluorescent microsphere aerosol challenge (3 conditions × 3 runs)	Pass	S4 P1 A
HS-04.1	H-05 (S4)	IS	IFU biosafety chapter: risk assessment, BSC or evacuation for infectious ...	SYS-554	DVP-504 TC-04; PVR-CSI-001	Fluorescent microsphere aerosol challenge (3 conditions × 3 runs)	Pass	S4 P1 A
HS-04.2	H-05 (S4)	PM	Clog detection < 500 ms; stream and HV off, drawer closed, tube unloaded within ...	SYS-543	DVP-503 TC-10; PVR-CSI-001	DVP-CSI-504 simulated clog; VAL-CSI-03; ANM-0011, ANM-0051	Pass (after ANM-0011)	S4 P1 A
HS-04.2	H-05 (S4)	PM	Clog detection < 500 ms; stream and HV off, drawer closed, tube unloaded within ...	FW-210	U: UTS-010-13, STS-B-014; SVR-SW-CSI-010	DVP-CSI-504 simulated clog; VAL-CSI-03; ANM-0011, ANM-0051	Pass	S4 P1 A

HS	Harm (S)	Type	Risk control measure	Req.	Verification of implementation	Effectiveness evidence	Result	Residual
HS-04.2	H-05 (S4)	PM	Clog detection < 500 ms; stream and HV off, drawer closed, tube unloaded within ...	FW-602	I: ITS §6, STS-B-031; SVR-SW-CSI-010	DVP-CSI-504 simulated clog; VAL-CSI-03; ANM-0011, ANM-0051	Pass (after ANM-0011)	S4 P1 A
HS-04.2	H-05 (S4)	PM	Guided recovery: boost evacuation before door-open step is shown	RA-504	S: STS-A-031; SVR-SW-CSI-010	DVP-CSI-504 simulated clog; VAL-CSI-03; ANM-0011, ANM-0051	Pass (after ANM-0051)	S4 P1 A
HS-04.2	H-05 (S4)	PM	Guided recovery: boost evacuation before door-open step is shown	RA-505	I: ITS §3, ITS §6, STS-A-032; SVR-SW-CSI-010	DVP-CSI-504 simulated clog; VAL-CSI-03; ANM-0011, ANM-0051	Pass	S4 P1 A
HS-04.2	H-05 (S4)	IS	IFU clog handling	SYS-554	DVP-504 TC-04; PVR-CSI-001	DVP-CSI-504 simulated clog; VAL-CSI-03; ANM-0011, ANM-0051	Pass	S4 P1 A
HS-04.3	H-05 (S4)	ID	Splash shield and chamber cover; pressurization only after port closed	RA-205	I: ITS §4, ITS §6, STS-A-010; SVR-SW-CSI-010	Seeded cracked-tube test	Pass (after ANM-0019)	S4 P1 A
HS-04.3	H-05 (S4)	IS	IFU: approved tube types only; PPE	SYS-1102	DVP-1101 TC-02; PVR-CSI-001	Seeded cracked-tube test	Pass	S4 P1 A
HS-04.4	H-05 (S4)	PM	Automated bleach clean cycle	SYS-603	DVP-601 TC-03; PVR-CSI-001	Residual protein test after clean cycle	Pass	S4 P1 A
HS-04.4	H-05 (S4)	PM	Automated bleach clean cycle	RA-203	S: STS-A-008; SVR-SW-CSI-010	Residual protein test after clean cycle	Pass	S4 P1 A
HS-04.4	H-05 (S4)	IS	Decontamination certificate required before service	SYS-554	DVP-504 TC-04; PVR-CSI-001	Residual protein test after clean cycle	Pass	S4 P1 A
HA-05: Access to sort chamber during sorting								
HS-05.1	H-05 (S4)	ID	Collection door cannot open unless sort block door is closed	SYS-521	DVP-503 TC-01; PVR-CSI-001	Door sequence test	Pass	S4 P1 A
HS-05.1	H-05 (S4)	PM	Firmware door monitoring	FW-502	I: ITS §6, STS-B-029; SVR-SW-CSI-010	Door sequence test	Pass	S4 P1 A
HS-05.1	H-05 (S4)	IS	IFU tube change	SYS-1102	DVP-1101 TC-02; PVR-CSI-001	Door sequence test	Pass	S4 P1 A
HA-06: Moving loading port								
HS-06.1	H-06 (S2)	ID	Force below IEC 61010-1 pinch limit	SYS-562	DVP-505 TC-02; PVR-CSI-001	Force measurement; stall test	Pass	S2 P1 A
HS-06.1	H-06 (S2)	PM	Illuminated e-stop; stall detection	SYS-531	DVP-503 TC-05; PVR-CSI-001	Force measurement; stall test	Pass	S2 P1 A
HS-06.1	H-06 (S2)	PM	Illuminated e-stop; stall detection	SYS-533	DVP-503 TC-07; PVR-CSI-001	Force measurement; stall test	Pass	S2 P1 A
HS-06.1	H-06 (S2)	PM	Illuminated e-stop; stall detection	FW-309	U: UTS-010-12, STS-B-023; SVR-SW-CSI-010	Force measurement; stall test	Pass	S2 P1 A
HS-06.1	H-06 (S2)	IS	Pinch label	SYS-1103	DVP-1101 TC-03; PVR-CSI-001	Force measurement; stall test	Pass	S2 P1 A
HS-06.2	H-06 (S2)	PM	Collection chamber door must be closed for stage motion; e-stop stops stage in ...	SYS-532	DVP-503 TC-06; PVR-CSI-001	E-stop timing	Pass	S2 P1 A
HS-06.2	H-06 (S2)	PM	Collection chamber door must be closed for stage motion; e-stop stops stage in ...	FW-601	I: ITS §6, STS-B-030; SVR-SW-CSI-010	E-stop timing	Pass	S2 P1 A
HS-06.2	H-06 (S2)	IS	IFU	SYS-1102	DVP-1101 TC-02; PVR-CSI-001	E-stop timing	Pass	S2 P1 A

HS	Harm (S)	Type	Risk control measure	Req.	Verification of implementation	Effectiveness evidence	Result	Residual
HA-07: Low purity of sorted fraction (breakoff drift)								
HS-07.1	H-10 (S4)	PM	Breakoff control; charging paused within 100 ms when out of range	SYS-613	DVP-601 TC-07; PVR-CSI-001	DVP-CSI-405 induced drift; VAL-CSI-02	Pass (after ANM-0014)	S4 P1 A
HS-07.1	H-10 (S4)	PM	Breakoff control; charging paused within 100 ms when out of range	SYS-542	DVP-503 TC-09; PVR-CSI-001	DVP-CSI-405 induced drift; VAL-CSI-02	Pass	S4 P1 A
HS-07.1	H-10 (S4)	PM	Breakoff control; charging paused within 100 ms when out of range	FW-205	I: ITS §7, ITS §9, STS-B-009; SVR-SW-CSI-010	DVP-CSI-405 induced drift; VAL-CSI-02	Pass	S4 P1 A
HS-07.1	H-10 (S4)	PM	Sort report with pauses and conflicts	RA-706	S: STS-A-043; SVR-SW-CSI-010	DVP-CSI-405 induced drift; VAL-CSI-02	Pass	S4 P1 A
HS-07.1	H-10 (S4)	IS	IFU: re-analyze sorted fraction	SYS-1102	DVP-1101 TC-02; PVR-CSI-001	DVP-CSI-405 induced drift; VAL-CSI-02	Pass	S4 P1 A
HS-07.2	H-10 (S4)	PM	Automatic drop delay with beads	SYS-465	DVP-405 TC-05; PVR-CSI-001	VAL-CSI-02	Pass	S4 P1 A
HS-07.2	H-10 (S4)	PM	Automatic drop delay with beads	RA-407	S: STS-A-027; SVR-SW-CSI-010	VAL-CSI-02	Pass	S4 P1 A
HS-07.2	H-10 (S4)	PM	Nozzle confirmation loads matching setup	RA-402	S: STS-A-022; SVR-SW-CSI-010	VAL-CSI-02	Pass	S4 P1 A
HS-07.3	H-10 (S4)	PM	Software advisory when sort population < 1%	SYS-633	DVP-601 TC-15; PVR-CSI-001	VAL-CSI-02 supplementary	Pass	S4 P2 R
HS-07.3	H-10 (S4)	IS	IFU event rate guidance	SYS-1102	DVP-1101 TC-02; PVR-CSI-001	VAL-CSI-02 supplementary	Pass	S4 P2 R
HA-08: Optical configuration mismatch								
HS-08.1	H-10 (S4)	ID	Configuration generated from controlled record with rules	AD-105	U: UTS-010-20, STS-C-005; SVR-SW-CSI-010	Seeded-error test; CAPA-2028-014 PA11 inventory	Pass (after ANM-0026)	S4 P2 R
HS-08.1	H-10 (S4)	ID	Configuration generated from controlled record with rules	AD-106	I: ITS §12, STS-C-006; SVR-SW-CSI-010	Seeded-error test; CAPA-2028-014 PA11 inventory	Pass	S4 P2 R
HS-08.1	H-10 (S4)	PM	Board inventory and configuration match check before acquisition	FW-302	I: ITS §6, ITS §8, STS-B-016; SVR-SW-CSI-010	Seeded-error test; CAPA-2028-014 PA11 inventory	Pass	S4 P2 R
HS-08.1	H-10 (S4)	PM	Board inventory and configuration match check before acquisition	RA-805	I: ITS §5, ITS §6, STS-A-048; SVR-SW-CSI-010	Seeded-error test; CAPA-2028-014 PA11 inventory	Pass	S4 P2 R
HS-08.1	H-10 (S4)	PM	New baseline required after change	AD-104	S: STS-C-004; SVR-SW-CSI-010	Seeded-error test; CAPA-2028-014 PA11 inventory	Pass	S4 P2 R
HS-08.2	H-10 (S4)	PM	Daily QC lockout	SYS-622	DVP-601 TC-10; PVR-CSI-001	CAPA effectiveness check E1 to E3	Pass (after ANM-0025)	S4 P2 R
HS-08.2	H-10 (S4)	PM	Daily QC lockout	RA-803	S: STS-A-046; SVR-SW-CSI-010	CAPA effectiveness check E1 to E3	Pass (after ANM-0025)	S4 P2 R
HS-08.2	H-10 (S4)	PM	Assay controls with every run; results flagged when warnings active	RA-903	S: STS-A-052; SVR-SW-CSI-010	CAPA effectiveness check E1 to E3	Pass	S4 P2 R
HS-08.2	H-10 (S4)	PM	Fan control on laser temperature; laser temperature warning (added Rev D)	FW-311 (D)	I: ITS §6, ITS §8, STS-B-053; SVR-SW-CSI-010 Add. 1	CAPA effectiveness check E1 to E3	Pass (SVR-SW Add. 1)	S4 P2 R
HS-08.2	H-10 (S4)	PM	Fan control on laser temperature; laser temperature warning (added Rev D)	RA-107 (D)	S: STS-A-062; SVR-SW-CSI-010 Add. 1	CAPA effectiveness check E1 to E3	Pass (SVR-SW Add. 1)	S4 P2 R

HS	Harm (S)	Type	Risk control measure	Req.	Verification of implementation	Effectiveness evidence	Result	Residual
HS-08.2	H-10 (S4)	PM	Thermal margin requirement (added Rev D)	SYS-327 (D)	DVP-301 TC-20; DVP-403 TC-09; PVR-CSI-001 Add. 2	CAPA effectiveness check E1 to E3	Pass (PVR Add. 2)	S4 P2 R
HS-08.3	H-10 (S4)	PM	QC lockout blocks IVD templates; UTC-based 24 h window	RA-803	S: STS-A-046; SVR-SW-CSI-010	Seeded tests incl. DST boundary	Pass (after ANM-0025)	S4 P1 A
HS-08.3	H-10 (S4)	PM	Override limited to research use and labeled	RA-804	S: STS-A-047; SVR-SW-CSI-010	Seeded tests incl. DST boundary	Pass	S4 P1 A
HS-08.4	H-10 (S4)	PM	Locked, signed assay templates; versioning	SYS-704	DVP-701 TC-04; PVR-CSI-001; SVR-SW-CSI-010	VAL-CSI-09	Pass	S4 P1 A
HS-08.4	H-10 (S4)	PM	Locked, signed assay templates; versioning	RA-901	S: STS-A-050; SVR-SW-CSI-010	VAL-CSI-09	Pass	S4 P1 A
HS-08.4	H-10 (S4)	PM	Locked, signed assay templates; versioning	AD-802	S: STS-C-035; SVR-SW-CSI-010	VAL-CSI-09	Pass	S4 P1 A
HS-08.4	H-10 (S4)	PM	Audit trail	DM-201	I: ITS §17, STS-D-006; SVR-SW-CSI-010	VAL-CSI-09	Pass (after ANM-0028)	S4 P1 A
HS-08.5	H-10 (S4)	PM	Compensation locked in IVD templates	RA-901	S: STS-A-050; SVR-SW-CSI-010	VAL-CSI-01	Pass	S4 P1 A
HS-08.5	H-10 (S4)	IS	IFU and assay IFU	SYS-1102	DVP-1101 TC-02; PVR-CSI-001	VAL-CSI-01	Pass	S4 P1 A
HS-08.6	H-13 (S4)	PM	Locked CD34 template with minimum event count	RA-901	S: STS-A-050; SVR-SW-CSI-010	VAL-CSI-01, VAL-CSI-04	Pass	S4 P2 R
HS-08.6	H-13 (S4)	PM	Acquisition rate and lost events reported	RA-304	I: ITS §2, STS-A-017; SVR-SW-CSI-010	VAL-CSI-01, VAL-CSI-04	Pass	S4 P2 R
HA-09: Pressurized sheath tank (up to 100 PSI supply)								
HS-09.1	H-09 (S3)	ID	Tanks rated 1.5× maximum with relief valves	SYS-561	DVP-505 TC-01; PVR-CSI-001	Relief valve test	Pass	S3 P1 A
HS-09.1	H-09 (S3)	PM	Pressure regulation and monitoring	FW-401	U: UTS-010-14, STS-B-025; SVR-SW-CSI-010	Relief valve test	Pass	S3 P1 A
HS-09.2	H-09 (S3)	PM	Venting on power off; refill procedure depressurizes	SYS-524	DVP-503 TC-04; PVR-CSI-001	Depressurization test	Pass	S3 P1 A
HS-09.2	H-09 (S3)	PM	Venting on power off; refill procedure depressurizes	FW-403	I: ITS §10, STS-B-027; SVR-SW-CSI-010	Depressurization test	Pass	S3 P1 A
HS-09.2	H-09 (S3)	IS	Tank label; IFU	SYS-1102	DVP-1101 TC-02; PVR-CSI-001	Depressurization test	Pass	S3 P1 A
HA-10: Decontamination agents								
HS-10.1	H-08 (S2)	ID	Compatible materials; drip tray	SYS-553	DVP-504 TC-03; PVR-CSI-001	Material test	Pass	S2 P1 A
HA-11: Instrument mass (185 kg)								
HS-11.1	H-07 (S3)	ID	Stable at 10° tilt	SYS-563	DVP-505 TC-03; PVR-CSI-001	Tilt test	Pass	S3 P1 A
HS-11.1	H-07 (S3)	IS	Installation by service only; site preparation guide	SYS-1104	DVP-1101 TC-04; PVR-CSI-001	Tilt test	Pass	S3 P1 A
HS-11.1	H-07 (S3)	IS	Installation by service only; site preparation guide	SYS-1203	DVP-1201 TC-03; PVR-CSI-001; ISTA report	Tilt test	Pass	S3 P1 A

HS	Harm (S)	Type	Risk control measure	Req.	Verification of implementation	Effectiveness evidence	Result	Residual
HA-12: Mains voltage								
HS-12.1	H-04 (S5)	ID	IEC 61010-1 double protection	SYS-501	DVP-501 TC-01; SVR-CSI-001	IEC 61010 tests	Pass	S5 P1 R
HS-12.1	H-04 (S5)	PM	100% ground bond, hipot, leakage in production	SYS-502	DVP-501 TC-02; SVR-CSI-001	IEC 61010 tests	Pass	S5 P1 R
HS-12.1	H-04 (S5)	PM	Circuit breaker	SYS-315	DVP-301 TC-11; PVR-CSI-001	IEC 61010 tests	Pass	S5 P1 R
HS-12.2	H-16 (S4)	ID	Flame-rated enclosure and materials	SYS-501	DVP-501 TC-01; SVR-CSI-001	Abnormal operation tests	Pass	S4 P1 A
HS-12.2	H-16 (S4)	PM	Temperature monitoring to safe state	FW-308	U: UTS-010-12, STS-B-022; SVR-SW-CSI-010	Abnormal operation tests	Pass	S4 P1 A
HS-12.2	H-16 (S4)	PM	Temperature monitoring to safe state	FW-606	I: ITS §6, ITS §8, STS-B-035; SVR-SW-CSI-010	Abnormal operation tests	Pass	S4 P1 A
HA-13: Electromagnetic disturbance								
HS-13.1	H-10 (S4)	ID	Shielded detector cabling; filters	SYS-503	DVP-501 TC-03; SVR-CSI-001	Immunity with monitoring	Pass (after ANM-0021)	S4 P1 A
HS-13.1	H-10 (S4)	PM	Daily QC	SYS-621	DVP-601 TC-09; PVR-CSI-001	Immunity with monitoring	Pass	S4 P1 A
HA-14: Fluid misconnection								
HS-14.1	H-12 (S3)	ID	Keyed non-interchangeable connectors; color coding	SYS-801	DVP-801 TC-01; PVR-CSI-001	Misconnection attempts	Pass	S3 P1 A
HS-14.1	H-12 (S3)	ID	Keyed non-interchangeable connectors; color coding	SYS-905	DVP-901 TC-05; PVR-CSI-001	Misconnection attempts	Pass	S3 P1 A
HA-15: Fluid run-out								
HS-15.1	H-12 (S3)	PM	Level monitoring; capacity check before sort	SYS-493	DVP-408 TC-03; PVR-CSI-001	Low-level scenarios	Pass	S3 P2 A
HS-15.1	H-12 (S3)	PM	Level monitoring; capacity check before sort	RA-207	S: STS-A-012; SVR-SW-CSI-010	Low-level scenarios	Pass	S3 P2 A
HS-15.1	H-12 (S3)	PM	Level monitoring; capacity check before sort	FW-402	I: ITS §10, ITS §6, STS-B-026; SVR-SW-CSI-010	Low-level scenarios	Pass	S3 P2 A
HS-15.2	H-12 (S3)	PM	Sample tube unloaded on clog (sample preserved)	FW-602	I: ITS §6, STS-B-031; SVR-SW-CSI-010	Clog test	Pass (after ANM-0011)	S3 P3 R
HS-15.2	H-12 (S3)	PM	Sort record preserved; resume	RA-703	I: ITS §3, STS-A-040; SVR-SW-CSI-010	Clog test	Pass	S3 P3 R
HS-15.2	H-12 (S3)	PM	Sort record preserved; resume	DM-104	S: STS-D-004; SVR-SW-CSI-010	Clog test	Pass	S3 P3 R
HA-16: Unauthorized modification								
HS-16.1	H-10 (S4)	ID	Isolated instrument link; locked image	SYS-707	DVP-701 TC-07; PVR-CSI-001; SVR-SW-CSI-010	Penetration test	Pass	S4 P1 A
HS-16.1	H-10 (S4)	ID	Isolated instrument link; locked image	CM-201	S: STS-E-004; SVR-SW-CSI-010	Penetration test	Pass	S4 P1 A
HS-16.1	H-10 (S4)	PM	Signed updates; pairing; session keys	FW-104	I: ITS §6, STS-B-004; SVR-SW-CSI-010	Penetration test	Pass	S4 P1 A

HS	Harm (S)	Type	Risk control measure	Req.	Verification of implementation	Effectiveness evidence	Result	Residual
HS-16.1	H-10 (S4)	PM	Signed updates; pairing; session keys	FW-802	I: ITS §6, ITS §19, STS-B-045; SVR-SW-CSI-010	Penetration test	Pass (after ANM-0030)	S4 P1 A
HS-16.1	H-10 (S4)	PM	Signed updates; pairing; session keys	FW-803	U: UTS-010-10, STS-B-046; SVR-SW-CSI-010	Penetration test	Pass (after ANM-0030)	S4 P1 A
HS-16.1	H-10 (S4)	PM	Audit trail	DM-201	I: ITS §17, STS-D-006; SVR-SW-CSI-010	Penetration test	Pass (after ANM-0028)	S4 P1 A
HS-16.2	H-15 (S2)	PM	Encryption at rest; access control	DM-504	I: ITS §16, STS-D-021; SVR-SW-CSI-010	Security test	Pass	S2 P1 A
HS-16.2	H-15 (S2)	PM	Encryption at rest; access control	CM-101	S: STS-E-001; SVR-SW-CSI-010	Security test	Pass	S2 P1 A
HS-16.2	H-15 (S2)	PM	De-identified export option	DM-403	S: STS-D-016; SVR-SW-CSI-010	Security test	Pass	S2 P1 A
HA-17: Specimen misidentification								
HS-17.1	H-10 (S4)	PM	Review and release with e-signature	DM-301	S: STS-D-010; SVR-SW-CSI-010	VAL-CSI-03, VAL-CSI-09	Pass	S4 P2 R
HS-17.1	H-10 (S4)	PM	Review and release with e-signature	SYS-703	DVP-701 TC-03; PVR-CSI-001; SVR-SW-CSI-010	VAL-CSI-03, VAL-CSI-09	Pass	S4 P2 R
HS-17.1	H-10 (S4)	PM	LIS export avoids transcription	DM-404	S: STS-D-017; SVR-SW-CSI-010	VAL-CSI-03, VAL-CSI-09	Pass (after ANM-0058)	S4 P2 R
HS-17.1	H-10 (S4)	IS	IFU and training	SYS-1102	DVP-1101 TC-02; PVR-CSI-001	VAL-CSI-03, VAL-CSI-09	Pass	S4 P2 R
HS-17.2	H-11 (S2)	PM	Fan control and thermal margin (CAPA-2028-014)	FW-311 (D)	I: ITS §6, ITS §8, STS-B-053; SVR-SW-CSI-010 Add. 1	Effectiveness check E1	Pass (SVR-SW Add. 1)	S2 P2 A
HS-17.2	H-11 (S2)	PM	Fan control and thermal margin (CAPA-2028-014)	SYS-327 (D)	DVP-301 TC-20; DVP-403 TC-09; PVR-CSI-001 Add. 2	Effectiveness check E1	Pass (PVR Add. 2)	S2 P2 A
HS-17.2	H-11 (S2)	PM	Laser temperature visible for faster troubleshooting	RA-107 (D)	S: STS-A-062; SVR-SW-CSI-010 Add. 1	Effectiveness check E1	Pass (SVR-SW Add. 1)	S2 P2 A

9. Matrix F: Standards and Regulations

Each applicable standard or regulation is traced to the system requirements that implement it and to the test report that shows compliance. Requirement lists are generated from SRS-CSI-001 source and section.

Standard / regulation	Scope	System requirements	Verified in	Evidence
IEC 61010-1:2010+A1; IEC 61010-2-101	Electrical, mechanical and fire safety; IVD particular requirements	501, 502, 561, 562, 563, 524	DVP-501, 505	SVR-CSI-001 (CB scheme)
IEC 61326-1; IEC 61326-2-6	EMC for IVD equipment	503	DVP-501	SVR-CSI-001
IEC 60825-1:2014	Laser product classification and labeling	511, 512, 513	DVP-502	SVR-CSI-001
ISO 14971:2019	Risk management; risk controls	1102, 1103, 1104, 1203, 315, 327, 436, 465, 493, 501, 502, 503, 511, 512, 513, 521, 522, 523, 524, 531, 532, 533, 542, 543, 551, 552, 553, 554, 561, 562, 563, 603, 613, 621, 622, 633, 703, 704, 707, 801, 804, 905, 1304	Matrix D	RMF-CSI-001 Rev D
IEC 62304 (A1); IEC 82304-1	Software life cycle	1305	STS, ITS, UTS	SVR-SW-CSI-010
IEC 62366-1 (A1)	Usability engineering	901, 902, 903, 904, 905	DVP-901; VAL-CSI-03	HFE-CSI-001
21 CFR Part 11	Electronic records and signatures	701, 702, 703, 709	DVP-701; VAL-CSI-09	SVR-SW-CSI-010
FD&C Act §524B; IEC 81001-5-1	Cybersecurity	707, 708, 709	DVP-701 (penetration test)	SVR-SW-CSI-010; PT-CSI-2027-01
ISO 15223-1; ISO 18113-3; 21 CFR 809.10	Symbols; IVD instrument information; IVD labeling	1101, 1102, 1103, 1104, 1105	DVP-1101	PVR-CSI-001
ISTA 3A	Transport simulation	1201	DVP-1201	ISTA report
21 CFR 820 (QMSR); ISO 13485:2016	Quality management system; design controls	1301, 1302	DVP-1301	DHF review
EU IVDR 2017/746 Annex I	General safety and performance requirements	1303	DVP-1301	GSPR checklist GSPR-CSI-001
RoHS 2011/65/EU; REACH	Restricted substances	1306	DVP-1301	Supplier declarations

IEC 62304 applies to all software requirements in Matrix C; SYS-1305 is the process requirement.

10. Matrix G: Information for Safety and Labeling

Information for safety is the lowest-priority control and is traced like any other: each IS control points to the labeling requirement and the labeling section that contains it. Effectiveness of critical labeling was shown in the summative usability study VAL-CSI-03.

HS	Information for safety	Labeling requirement	Where in labeling	Effectiveness
HS-01.1	Warning labels; IFU	SYS-513, SYS-1103	OG-CSI-001 §2.3	DVP-1101 inspection

HS	Information for safety	Labeling requirement	Where in labeling	Effectiveness
HS-01.2	Service manual laser safety procedure, eyewear by wavelength, training	SYS-1104	SM-CSI-001 §3	DVP-1101 inspection
HS-01.3	IFU	SYS-1102	OG-CSI-001; LBL-CSI-001	DVP-1101 inspection
HS-02.1	HV label at sort block; IFU cleaning procedure requires HV off	SYS-1103	OG-CSI-001 §6.4	VAL-CSI-03 T-04: 0 of 30
HS-04.1	IFU biosafety chapter: risk assessment, BSC or evacuation for infectious specimens	SYS-554	OG-CSI-001 §3	DVP-1101 inspection
HS-04.2	IFU clog handling	SYS-554	OG-CSI-001 §7.2	VAL-CSI-03 T-02: 0 of 30 (after ANM-0051)
HS-04.3	IFU: approved tube types only; PPE	SYS-1102	OG-CSI-001 §5.1	DVP-1101 inspection
HS-04.4	Decontamination certificate required before service	SYS-554	SM-CSI-001 §2	VAL-CSI-03 T-10: 0 of 30
HS-05.1	IFU tube change	SYS-1102	OG-CSI-001 §7.1	VAL-CSI-03 T-03: 0 of 30
HS-06.1	Pinch label	SYS-1103	Label	VAL-CSI-03 T-01: 0 of 30
HS-06.2	IFU	SYS-1102	OG-CSI-001; LBL-CSI-001	DVP-1101 inspection
HS-07.1	IFU: re-analyze sorted fraction	SYS-1102	OG-CSI-001 §7.4	DVP-1101 inspection
HS-07.3	IFU event rate guidance	SYS-1102	OG-CSI-001 §7.4	VAL-CSI-03 T-12: 0 of 30 (supplementary)
HS-08.5	IFU and assay IFU	SYS-1102	OG-CSI-001; LBL-CSI-001	DVP-1101 inspection
HS-09.2	Tank label; IFU	SYS-1102	OG-CSI-001 §5.3	DVP-1101 inspection
HS-11.1	Installation by service only; site preparation guide	SYS-1104, SYS-1203	SPG-CSI-001	DVP-1101 inspection
HS-17.1	IFU and training	SYS-1102	OG-CSI-001 §10	VAL-CSI-03 T-11: 1 of 30 (caught at review)

11. Backward Trace

Every test, design output and production control must be justified by a requirement. These tables trace each back to what it verifies or implements.

11.1 System verification protocols and test cases to system requirements

Protocol	Title	Test cases	System requirements verified	Report
DVP-CSI-301	Physical, electrical and environmental	20	301, 302, 303, 304, 305, 306, 311, 312, 313, 314, 315, 316, 321, 322, 323, 324, 325, 326, 327 (D), 331	PVR-CSI-001 Add. 2
DVP-CSI-401	Fluorescence resolution and linearity	3	401, 402, 411	PVR-CSI-001
DVP-CSI-402	Scatter sensitivity and detection	8	421, 422, 423, 441, 442, 443, 444, 445	PVR-CSI-001
DVP-CSI-403	Excitation (lasers, beam geometry, warm-up)	9	431, 432, 433, 434, 435, 436, 437, 438, 327 (D)	PVR-CSI-001 Add. 2
DVP-CSI-404	Acquisition rate and sample flow	3	451, 452, 453	PVR-CSI-001
DVP-CSI-405	Sort performance (purity, yield, viability, nozzles)	8	461, 462, 463, 464, 465, 466, 467, 468	PVR-CSI-001
DVP-CSI-406	Carryover	2	471, 472	PVR-CSI-001
DVP-CSI-407	Sort collection and plate deposition	5	481, 482, 483, 484, 485	PVR-CSI-001
DVP-CSI-408	Fluidics and temperature control	7	491, 492, 493, 494, 495, 4101, 4102	PVR-CSI-001
DVP-CSI-501	Electrical safety and EMC	3	501, 502, 503	SVR-CSI-001
DVP-CSI-502	Laser safety	3	511, 512, 513	SVR-CSI-001
DVP-CSI-503	Interlocks, HV, e-stop, clog and breakoff response	10	521, 522, 523, 524, 531, 532, 533, 541, 542, 543	PVR-CSI-001
DVP-CSI-504	Biohazard containment and decontamination	4	551, 552, 553, 554	PVR-CSI-001
DVP-CSI-505	Mechanical and pressure safety	3	561, 562, 563	PVR-CSI-001
DVP-CSI-601	Functional requirements	17	601, 602, 603, 604, 611, 612, 613, 614, 621, 622, 623, 624, 631, 632, 633, 634, 635	PVR-CSI-001
DVP-CSI-701	Software, data integrity and cybersecurity	9	701, 702, 703, 704, 705, 706, 707, 708, 709	PVR-CSI-001; SVR-SW-CSI-010
DVP-CSI-801	Interfaces	6	801, 802, 803, 804, 805, 806	PVR-CSI-001
DVP-CSI-901	Usability requirements (design verification)	5	901, 902, 903, 904, 905	PVR-CSI-001
DVP-CSI-1001	Reliability, maintainability and serviceability	8	1001, 1002, 1003, 1004, 1005, 1006, 1007, 1008	PVR-CSI-001
DVP-CSI-1101	Labeling and documentation	5	1101, 1102, 1103, 1104, 1105	PVR-CSI-001
DVP-CSI-1201	Packaging, transport and installation	4	1201, 1202, 1203, 1204	PVR-CSI-001; ISTA report

Protocol	Title	Test cases	System requirements verified	Report
DVP-CSI-1301	Regulatory and standards compliance	6	1301, 1302, 1303, 1304, 1305, 1306	PVR-CSI-001

148 system test cases in 22 protocols. DVP-CSI-301 and DVP-CSI-403 were repeated in 2028 with lasers from two supplier lots (VVP-CSI-001 Rev B; PVR-CSI-001 Add. 2).

11.2 Software test specifications to software requirements

Test specification	Scope	Software requirements
UTS-CSI-010-02	Unit tests, SI-02	RA-302, RA-305
UTS-CSI-010-03	Unit tests, SI-03	RA-603, RA-605
UTS-CSI-010-10	Unit tests, SI-10	FW-605, FW-702, FW-704, FW-706, FW-803, FW-804
UTS-CSI-010-11	Unit tests, SI-11	FW-102, FW-201, FW-202, FW-203, FW-204, FW-206
UTS-CSI-010-12	Unit tests, SI-12	FW-102, FW-303, FW-304, FW-305, FW-306, FW-307, FW-308, FW-309
UTS-CSI-010-13	Unit tests, SI-13	FW-209, FW-210
UTS-CSI-010-14	Unit tests, SI-14	FW-401
UTS-CSI-010-15	Unit tests, SI-15	FW-604, FW-605
UTS-CSI-010-20	Unit tests, SI-20	AD-105
UTS-CSI-010-22	Unit tests, SI-22	AD-602
UTS-CSI-010-30	Unit tests, SI-30	DM-401, DM-402
UTS-CSI-010-31	Unit tests, SI-31	DM-103
UTS-CSI-010-32	Unit tests, SI-32	DM-202, DM-303
UTS-CSI-010-40	Unit tests, SI-40	CM-103
ITS-CSI-010 §1	Integration, SI-01	RA-1003, CM-305
ITS-CSI-010 §2	Integration, SI-02	RA-304, RA-307
ITS-CSI-010 §3	Integration, SI-03	RA-403, RA-404, RA-501, RA-502, RA-503, RA-505, RA-703, RA-705
ITS-CSI-010 §4	Integration, SI-04	RA-204, RA-205
ITS-CSI-010 §5	Integration, SI-05	RA-802, RA-805, RA-904
ITS-CSI-010 §6	Integration, SI-10	RA-204, RA-205, RA-505, RA-805, FW-101, FW-103, FW-104, FW-208, FW-302, FW-311 (D), FW-312 (D), FW-313 (D), FW-402, FW-501, FW-502, FW-601, FW-602, FW-603, FW-606, FW-607, FW-608, FW-701, FW-703, FW-705, FW-801, FW-802, FW-805, FW-806, FW-807, FW-808, AD-102
ITS-CSI-010 §7	Integration, SI-11	RA-802, FW-205, FW-207, FW-806
ITS-CSI-010 §8	Integration, SI-12	FW-301, FW-302, FW-310, FW-311 (D), FW-606, AD-502
ITS-CSI-010 §9	Integration, SI-13	RA-404, RA-501, FW-205, AD-501
ITS-CSI-010 §10	Integration, SI-14	FW-402, FW-403, AD-401, AD-403
ITS-CSI-010 §11	Integration, SI-15	FW-103
ITS-CSI-010 §12	Integration, SI-20	AD-102, AD-106
ITS-CSI-010 §13	Integration, SI-21	AD-401, AD-403, AD-501, AD-502
ITS-CSI-010 §15	Integration, SI-30	CM-202
ITS-CSI-010 §16	Integration, SI-31	RA-307, RA-705, RA-904, DM-504

Test specification	Scope	Software requirements
ITS-CSI-010 §17	Integration, SI-32	DM-201
ITS-CSI-010 §19	Integration, SI-41	FW-801, FW-802, FW-805, FW-806, FW-807, FW-808
STS-CSI-010 §A	System test, Research Application	62 test cases STS-A-001 to STS-A-062
STS-CSI-010 §B	System test, Firmware	55 test cases STS-B-001 to STS-B-055
STS-CSI-010 §C	System test, Admin Application	40 test cases STS-C-001 to STS-C-040
STS-CSI-010 §D	System test, Data Management Application	23 test cases STS-D-001 to STS-D-023
STS-CSI-010 §E	System test, Common (security and communication)	4 test cases STS-E-001 to STS-E-004

11.3 Validation studies to user needs

Study	Title	User needs	Result
VAL-CSI-01	Method comparison	UN-01, UN-03	Pass
VAL-CSI-02	Clinical sort purity, recovery and viability	UN-02, UN-05, UN-15	Pass (after ANM-0056)
VAL-CSI-03	Summative usability (human factors validation)	UN-06, UN-09	Pass (after ANM-0060)
VAL-CSI-04	Multi-site precision	UN-08, UN-01	Pass
VAL-CSI-05	Laboratory workflow and throughput	UN-04	Pass
VAL-CSI-06	Installation in clinical laboratories	UN-07	Pass
VAL-CSI-07	Carryover with clinical specimens	UN-14	Pass
VAL-CSI-08	Temperature control with cells	UN-11	Pass
VAL-CSI-09	Records, review and release in use	UN-10	Pass (after ANM-0058)
VAL-CSI-10	Serviceability in the field	UN-12	Pass
VAL-CSI-11	Configuration and field upgrade	UN-16	Pass
VAL-CSI-12	Regulatory and labeling review	UN-13	Pass

11.4 Software items to requirements

Item	Name	Units	Software requirements implemented
SI-01	Operator UI and Workflow	5	RA-101, RA-102, RA-103, RA-104, RA-105, RA-107 (D), RA-504, RA-702, RA-1001, RA-1002, RA-1003, CM-301, CM-302, CM-305, CM-401
SI-02	Acquisition and Analysis	5	RA-301, RA-302, RA-303, RA-304, RA-305, RA-306, RA-307, RA-901, RA-903, CM-303
SI-03	Sort Control	6	RA-401, RA-402, RA-403, RA-404, RA-405, RA-406, RA-407, RA-501, RA-502, RA-503, RA-505, RA-601, RA-602, RA-603, RA-604, RA-605, RA-701, RA-702, RA-703, RA-704, RA-705, RA-706
SI-04	Fluidics and Instrument Control	4	RA-201, RA-202, RA-203, RA-204, RA-205, RA-206, RA-207, RA-208, RA-1002
SI-05	QC Run and Assay Execution	4	RA-107 (D), RA-701, RA-801, RA-802, RA-803, RA-804, RA-805, RA-806, RA-901, RA-902, RA-903, RA-904, AD-104, AD-203, AD-903

Item	Name	Units	Software requirements implemented
SI-10	Instrument Controller Firmware	10	RA-204, RA-205, RA-505, RA-805, FW-101, FW-103, FW-104, FW-208, FW-302, FW-311 (D), FW-312 (D), FW-313 (D), FW-402, FW-501, FW-502, FW-601, FW-602, FW-603, FW-605, FW-606, FW-607, FW-608, FW-701, FW-702, FW-703, FW-704, FW-705, FW-706, FW-801, FW-802, FW-803, FW-804, FW-805, FW-806, FW-807, FW-808, AD-102, CM-203, CM-303
SI-11	Sort Logic FPGA	8	RA-802, FW-102, FW-201, FW-202, FW-203, FW-204, FW-205, FW-206, FW-207, FW-806
SI-12	Board Management FPGA	6	FW-102, FW-301, FW-302, FW-303, FW-304, FW-305, FW-306, FW-307, FW-308, FW-309, FW-310, FW-311 (D), FW-606, AD-502, AD-504
SI-13	Stream Imaging Firmware	4	RA-404, RA-407, RA-501, FW-205, FW-209, FW-210, AD-501
SI-14	Fluidics Module Firmware	4	FW-401, FW-402, FW-403, AD-401, AD-403
SI-15	Safety Supervisor Firmware	4	FW-103, FW-604, FW-605
SI-20	Hardware Configuration Control	4	AD-101, AD-102, AD-103, AD-104, AD-105, AD-106, AD-107, AD-305
SI-21	Measurement and Calibration Tools	6	AD-201, AD-202, AD-203, AD-204, AD-205, AD-301, AD-302, AD-303, AD-304, AD-305, AD-401, AD-402, AD-403, AD-404, AD-405, AD-501, AD-502, AD-503, AD-504, AD-505, AD-601, AD-901, AD-902, AD-903, AD-904, AD-905
SI-22	User, Assay and Bead Lot Administration	5	AD-601, AD-602, AD-603, AD-701, AD-702, AD-703, AD-801, AD-802
SI-30	Data Management Application	5	DM-101, DM-102, DM-105, DM-203, DM-301, DM-304, DM-401, DM-402, DM-403, DM-404, DM-501, DM-502, DM-503, CM-202, CM-203, CM-304
SI-31	Database and Records Service	5	RA-307, RA-705, RA-706, RA-904, AD-902, DM-103, DM-104, DM-105, DM-504
SI-32	Audit Trail and E-Signature Service	3	AD-103, AD-701, AD-802, DM-201, DM-202, DM-204, DM-302, DM-303, DM-304, DM-503
SI-40	Security Service	4	RA-804, AD-801, DM-204, CM-101, CM-102, CM-103
SI-41	Instrument Communication Library	5	RA-303, FW-801, FW-802, FW-805, FW-806, FW-807, FW-808, CM-201, CM-302, CM-401

11.5 Design outputs to system requirements

Design output	System requirements implemented
DHF-CSI-000	5: 1301, 1302, 1303, 1304, 1306
DWG-CSI-0001	45: 301, 302, 305, 312, 313, 316, 321, 322, 323, 324, 331, 453, 466, 471, 472, 481, 482, 483, 494, 495, 4101, 4102, 501, 502, 503, 511, 513, 551, 552, 553, 554, 562, 563, 604, 803, 804, 806, 901, 902, 903, 1001, 1002, 1003, 1005, 1006
DWG-CSI-1000, DWG-CSI-1100	19: 303, 304, 306, 314, 325, 326, 491, 492, 493, 524, 561, 601, 602, 603, 801, 803, 805, 905, 1004
DWG-CSI-3100	5: 401, 402, 467, 468, 903
DWG-CSI-4000	15: 327 (D), 401, 402, 421, 422, 431, 432, 433, 434, 435, 436, 437, 438, 512, 1008
DWG-CSI-5100, DWG-CSI-5200	10: 327 (D), 401, 402, 411, 421, 422, 423, 441, 442, 1008
DWG-CSI-7000	16: 461, 462, 463, 465, 467, 483, 521, 522, 523, 541, 542, 543, 611, 612, 613, 614
LBL-CSI-001	5: 1101, 1102, 1103, 1104, 1105
OG-CSI-001	5: 1101, 1102, 1103, 1104, 1105

Design output	System requirements implemented
PKG-CSI-001	4: 1201, 1202, 1203, 1204
SAD-SW-CSI-011	39: 327 (D), 444, 464, 465, 472, 484, 485, 493, 522, 532, 541, 542, 543, 601, 602, 603, 611, 621, 622, 623, 624, 631, 632, 633, 634, 635, 701, 702, 703, 704, 705, 706, 707, 708, 709, 802, 904, 1007, 1305
SCH-CSI-6000	37: 301, 302, 305, 311, 312, 313, 314, 315, 316, 321, 322, 323, 324, 327 (D), 331, 411, 435, 443, 444, 445, 451, 452, 461, 464, 501, 502, 503, 511, 513, 531, 532, 533, 553, 554, 563, 802, 1007
SPG-CSI-001	4: 1201, 1202, 1203, 1204
SSD-CSI-100	27: 303, 304, 306, 314, 325, 326, 453, 471, 472, 491, 492, 493, 494, 495, 4101, 524, 561, 562, 601, 602, 603, 801, 803, 805, 903, 905, 1004
SSD-CSI-300	5: 401, 402, 467, 468, 903
SSD-CSI-400	15: 327 (D), 401, 402, 421, 422, 431, 432, 433, 434, 435, 436, 437, 438, 512, 1008
SSD-CSI-500	10: 327 (D), 401, 402, 411, 421, 422, 423, 441, 442, 1008
SSD-CSI-600	18: 311, 314, 315, 327 (D), 411, 435, 443, 444, 445, 451, 452, 461, 464, 531, 532, 533, 802, 1007
SSD-CSI-700	16: 461, 462, 463, 465, 467, 483, 521, 522, 523, 541, 542, 543, 611, 612, 613, 614
SSD-CSI-800	7: 481, 482, 483, 4102, 562, 804, 903
SSD-CSI-900	3: 551, 552, 803

12. Anomaly Links

Every anomaly in ADL-CSI-001 is linked to the requirement or user need it affected and, where it relates to a hazard, to the hazardous situations of that hazard. Status is as of this revision.

Anomaly	Title	Requirements	Sev.	Hazardous situations	Status
ANM-0003	Sort decision latency exceeds 20 μ s above 95,000 events/s	FW-204, SYS-461	2	07.1, 07.2, 07.3	Closed
ANM-0007	Safety supervisor removes HV enable during normal boot	FW-604, FW-103	3	02.1, 02.2, 02.3	Closed
ANM-0011	Clog response exceeds 2 s in 2 of 59 trials	SYS-543, FW-602	1	04.1, 04.2, 04.3, 04.4	Closed
ANM-0012	PI CV 3.1% on DV-02 on day 2	SYS-401	2	08.1, 08.2, 08.3, 08.4, 08.5, 08.6	Closed
ANM-0014	Breakoff control oscillates with 130 μ m nozzle	RA-501, SYS-613	3	—	Closed
ANM-0016	Laser calibration factor applied to wrong laser on shared beam spot	AD-202	2	08.1, 08.2, 08.3, 08.4, 08.5, 08.6	Closed
ANM-0019	Carryover 0.7% with 1 mL microtube holder	SYS-471, RA-205	2	—	Closed
ANM-0021	Radiated immunity: event rate spikes at 3 V/m, 80 to 120 MHz	SYS-503	2	13.1	Closed
ANM-0023	After link loss, Research App shows Ready while firmware is in safe state	FW-805, RA-1002	2	02.1, 02.2, 02.3	Closed
ANM-0025	QC lockout window 25 h across daylight saving change	RA-803, SYS-622	2	08.1, 08.2, 08.3, 08.4, 08.5, 08.6	Closed
ANM-0026	480/20 filter rule not enforced after field upgrade to 445 nm	AD-105	2	08.1, 08.2, 08.3, 08.4, 08.5, 08.6	Closed
ANM-0028	Audit trail missing old value for gate edits in draft assay template	DM-201	2	—	Closed
ANM-0030	Replayed command accepted after reconnect (penetration test)	FW-802, FW-803	2	08.1, 08.2, 08.3, 08.4, 08.5, 08.6	Closed
ANM-0032	Laser warning label missing on rear panel of 5-laser build	SYS-513	2	01.1, 01.2, 01.3	Closed
ANM-0034	PMT HV readback fault at 1,000 V	FW-304	3	—	Closed
ANM-0035	Internal log loses last entries on power loss	FW-703	3	—	Closed
ANM-0037	384-well plate edge wells off target	SYS-483, AD-504	3	—	Closed
ANM-0039	Sample chamber takes 52 min to reach 4 °C at 30 °C ambient	SYS-4101	3	—	Closed
ANM-0040	DV-05 needed realignment after ISTA 3E transport test	SYS-1201	2	08.1, 08.2, 08.3, 08.4, 08.5, 08.6	Closed
ANM-0042	Idle session lock not applied during calibration wizard	CM-103	3	—	Closed
ANM-0044	Memory growth in plotting library after 20 h acquisition	CM-303	3	—	Closed
ANM-0045	FCS export missing \$PnV keyword for FSC	DM-401	3	—	Closed
ANM-0047	Test failure: purity 96.8% on DV-01	SYS-461	3	—	Closed - not a defect

Anomaly	Title	Requirements	Sev.	Hazardous situations	Status
ANM-0048	Duplicate of ANM-0023	RA-1002	2	02.1, 02.2, 02.3	Closed - duplicate
ANM-0051	Usability verification: clog recovery step skipped	SYS-902, RA-504	3	04.1, 04.2, 04.3, 04.4	Closed
ANM-0053	German UI text truncated in sort layout editor	CM-305	4	—	Deferred
ANM-0056	VAL-CSI-02 Site 3: CD34+ purity 93% on two specimens	UN-02	2	07.1, 07.2, 07.3	Closed
ANM-0058	VAL-CSI-09 Site 2: LIS export rejected by site LIS	UN-10, DM-404	3	—	Closed
ANM-0060	VAL-CSI-03: users unsure which nozzle is installed	UN-09	3	08.1, 08.2, 08.3, 08.4, 08.5, 08.6	Closed
ANM-0062	Precision study: Site 1 day 3 QC warning on violet array	UN-08	3	—	Closed (CAPA-2028-014)
ANM-0064	Data Management search slow above 50,000 records	UN-10	4	—	Deferred
ANM-0066	Plate voltage readback drift warning after 6 h sorting	FW-307	3	02.1, 02.2, 02.3	Closed (board FPGA 3.0.6, Sep 2027)

13. Matrix E: Critical Characteristics to Production Controls

Design transfer extension (DTP-CSI-001 Section 6): each critical characteristic tested in production traces from its design requirement to the production control. The last two rows were added by CAPA-2028-014 (ECO-0241, ECO-0244).

Characteristic	Production limit	Design requirement	Production control
Fluorescence resolution (PI)	$CV \leq 2.7\%$	SRS-CSI-001 §4.1	FTS-CSI-001 §5.1
Fluorescence resolution (Hoechst)	$CV \leq 3.2\%$	SRS-CSI-001 §4.1	FTS-CSI-001 §5.2
Linearity	1.97–2.03	SRS-CSI-001 §4.2	FTS-CSI-001 §5.3
Scatter sensitivity	Resolved with ND 1.0 removed	SRS-CSI-001 §4.3	FTS-CSI-001 §5.4
Acquisition rate	No data loss at 75,000 events/s	SRS-CSI-001 §4.6	FTS-CSI-001 §5.5
Performance QC baseline	Pass; baseline stored in DHR	SRS-SW-CSI-010 §6.6	FTS-CSI-001 §5.6
Breakoff stability	≤ 5 px drift; Gap within target +1	SSD-CSI-700 §3.3	FTS-CSI-001 §5.7
Drop delay	Within ± 0.25 drop of manual method	SSD-CSI-700 §3.4	FTS-CSI-001 §5.8
Sort purity / yield	$\geq 99\%$ / $\geq 85\%$	SRS-CSI-001 §4.7	FTS-CSI-001 §5.9
Plate deposition (option)	$\geq 98\%$ of wells with exactly one bead	SSD-CSI-800 §4	FTS-CSI-001 §5.10
Clog response	All four actions within 1 s	SRS-CSI-001 §5.5	FTS-CSI-001 §5.11
Containment (with aerosol option)	No particles detected outside the sort chamber	SRS-CSI-001 §5.6	FTS-CSI-001 §5.12
Evacuator flow	≥ 7.5 / ≥ 32 CFM	SSD-CSI-900 §3.2	FTS-CSI-001 §5.13
Sample chamber temperature	± 1 °C at 45 min	SSD-CSI-100 §7.2	FTS-CSI-001 §5.14
Fluidics pressure	84–96 PSI	SSD-CSI-100 §4.2	FTS-CSI-001 §5.15

Characteristic	Production limit	Design requirement	Production control
Electrical safety	100% pass	SRS-CSI-001 §5.1	FTS-CSI-001 §5.16
Acoustic noise	≤ 60 dBA	SRS-CSI-001 §3.4	FTS-CSI-001 §5.17
E-stop and interlocks	100% pass	SRS-CSI-001 §5.4	FTS-CSI-001 §5.18
Thermal stability after 6 h soak at 30 °C room (all lasers on)	Violet and all-detector rCV and intensity within daily QC limits	SYS-327 (D)	FTS-CSI-001 Rev B §5.19
Laser module firmware version and 2 h hot test (incoming)	Approved version (AVL-CSI-001); power drift $\leq 2\%$ at 32 °C base	SYS-327 (D)	INS-CSI-001 Rev B §4

14. Change Impact: CAPA-2028-014

When a requirement, design output or test changes, the requirements tool marks every link to it as suspect. A suspect link is cleared only after an engineer records whether the linked item is affected and what evidence shows it still holds. This section records the change impact analysis for CAPA-2028-014 (violet detector QC failures from laser thermal drift), which is the only design change since Rev C.

14.1 Items added or changed

Item	Change	Traced up to	Traced down to	Evidence
SYS-327 (new)	Thermal margin requirement	UN-07; HA-08 (HS-08.2, HS-17.2)	SR-400.15; SR-500.10; SR-600.18; FW-311, FW-312, RA-107	DVP-301 TC-20; DVP-403 TC-09; PVR-CSI-001 Add. 2
FW-311 (new)	SI-10 shall control fan duty from the hottest laser base temperature ...	SYS-327, HA-08	SI-10, SI-12; ITS §6, ITS §8, STS-B-053	SVR-SW-CSI-010 Add. 1
FW-312 (new)	SI-10 shall log each laser base temperature once per minute and every ...	SYS-1007, SYS-327	SI-10; ITS §6, STS-B-054	SVR-SW-CSI-010 Add. 1
FW-313 (new)	SI-10 shall report the firmware version of every purchased module ...	SYS-1007, SYS-1008	SI-10; ITS §6, STS-B-055	SVR-SW-CSI-010 Add. 1
RA-107 (new)	The application shall show laser base temperatures in the status ...	SYS-327, SYS-622, HA-08	SI-01, SI-05; STS-A-062	SVR-SW-CSI-010 Add. 1
FTS-CSI-001 Rev B §5.19	6 h thermal soak added to factory acceptance	SYS-327	WI-CSI-1010 Rev C	TMV addendum 4; ECO-0241
INS-CSI-001 Rev B §4	Laser firmware check and hot test at incoming	SYS-327	Hot-test fixture qualification	ECO-0244
BOM-CSI-0001 Rev D; AVL-CSI-001 Rev C	Approved firmware as controlled attribute of purchased parts	SYS-1008	PLM rule	ECO-0246
VVP-CSI-001 Rev B	Temperature-sensitive verification uses two supplier lots	SYS-321, SYS-327	DVP-CSI-301, DVP-CSI-403	PVR-CSI-001 Add. 2
RMF-CSI-001 Rev D	HS-08.2 P1 revised; HS-17.2 added; new controls	HA-08, HA-17	Matrix D rows for HS-08.2, HS-17.2	RMR-CSI-001 Rev B

14.2 Suspect links raised and cleared

Suspect because	Links	Assessment	Evidence	Status
Controller firmware 3.0.2 changed (SI-10, SI-12)	48 software requirements on SI-10 or SI-12	Fan control and logging added; other behavior unchanged. Full regression of items rather than selective retest.	STS §B, ITS §6 and §8 full regression; UTS-010-10, -12	Cleared
Research Application 3.0.2 changed (SI-01, SI-05)	28 software requirements on SI-01 or SI-05	Status panel and QC report changed; QC logic unchanged	STS §A regression; formative check 5 of 5 users	Cleared
Thermal behavior of lasers changed	9 system requirements (SYS-321, SYS-431 to 438)	Performance at temperature re-verified with two laser lots	PVR-CSI-001 Add. 2 (lowest margin 2.6 °C)	Cleared
HS-08.2 estimate changed	12 risk control links (Matrix D, HS-08.2 and HS-17.2)	Controls re-verified; residual re-evaluated	RMF-CSI-001 Rev D §6, §8.2	Cleared
UN-08 validation evidence	VAL-CSI-04	Precision unaffected by fan control; ANM-0062 cause removed	Post-CAPA field data: 0 thermal QC failures in 5,412 runs	Cleared
Production controls for lasers	Matrix E rows for excitation	Two rows added; existing limits unchanged	FTS-CSI-001 Rev B	Cleared

No suspect links remain open at this revision.

15. Submission Map

Where the evidence traced in this matrix is presented in regulatory submissions. The 510(k) was cleared in 2027 on Rev C; Rev D supports the letter-to-file assessment for CAPA-2028-014 (LTF-2028-06: firmware and manufacturing changes that do not affect safety or effectiveness beyond the cleared device, no new 510(k) required) and the EU technical documentation update.

Submission section	What the reviewer checks	Matrices	Source records
510(k) eSTAR: Device description	Intended use, principle of operation, configurations	A, A2	SRS-CSI-001 §2; CFG-CSI-001
510(k) eSTAR: Software (Enhanced documentation level)	Requirements, architecture, V&V, traceability, anomalies	C, 11.2, 11.4, 12	SRS-SW-CSI-010; SAD-SW-CSI-011; SVR-SW-CSI-010; ADL-CSI-001
510(k) eSTAR: Cybersecurity (§524B)	Threat model, controls, SBOM, testing	D (HS-16.x), F	TM-SEC-CSI-001; PT-CSI-2027-01
510(k) eSTAR: Electrical safety and EMC	Standards and test reports	F	SVR-CSI-001
510(k) eSTAR: Performance testing (analytical)	Protocols, acceptance criteria, results	B, 11.1	PVR-CSI-001; DVAL-CSI-001
510(k) eSTAR: Human factors	Critical tasks, summative results	G, A (UN-06, UN-09)	HFE-CSI-001; VAL-CSI-03
510(k) eSTAR: Labeling	IFU, labels, symbols	G, F	OG-CSI-001; LBL-CSI-001
510(k) eSTAR: Risk management	Hazard analysis, controls, residual risk	D	RMF-CSI-001
EU IVDR Annex II §4 and §6	GSPR conformity; verification and validation	F, B, A	GSPR-CSI-001; this matrix

Submission section	What the reviewer checks	Matrices	Source records
EU IVDR Annex III	Post-market surveillance feedback into design	14	PMS-CSI-001; CAPA-2028-014

16. Reference Documents

Document no.	Rev	Title
SRS-CSI-001	E	System Requirements Specification
SSD-CSI-100 to SSD-CSI-950	A to C	Subsystem Design Specifications
SSVR-CSI-100 to SSVR-CSI-900	A	Subsystem Verification Reports
SRS-SW-CSI-010	F	Software Requirements Specification
SAD-SW-CSI-011	C	Software Architecture Description
SDS-SW-CSI-012 / SDS-FW-CSI-013	C / C	Software / Firmware Detailed Design Specifications
VVP-CSI-001	B	System Verification and Validation Plan
PVR-CSI-001 (Add. 1, 2) / SVR-CSI-001	A / A	System Performance Verification Report / Safety, EMC and Laser Report
SVR-SW-CSI-010 (Add. 1)	A	Software Verification Report
DVAL-CSI-001	A	Design Validation Report
RMF-CSI-001	D	Risk Management File (includes HA-CSI-001 Rev D, RMR-CSI-001 Rev B)
HFE-CSI-001	B	Human Factors Engineering Report
ADL-CSI-001	Export 11	Anomaly and Defect Tracking Log
CAPA-2028-014	Closed	CAPA Report: violet detector QC failures
DTP-CSI-001 / FTS-CSI-001 / INS-CSI-001	B / B / B	Design Transfer Plan / Factory Alignment and Test Specification / Incoming Inspection
BOM-CSI-0001 / AVL-CSI-001	D / C	Bill of Materials / Approved Vendor List
OG-CSI-001 / LBL-CSI-001	D / C	Operator's Guide / Labeling Specification
GSPR-CSI-001 / LTF-2028-06	B / A	GSPR Checklist / Letter-to-File for CAPA-2028-014 changes
VVR-CSI-001	A	V&V Summary Report